

```

clear; clc;

%=====
%                CARGAMOS LOS DATOS OBTENIDOS DEL BITALINO
%=====

% Seleccionar la carpeta donde están los archivos
carpeta = 'C:\ADRIANO ESTRADA\Intro_senales_biomedicas\PROYECTO_SENIALES';

% Buscar todos los archivos .txt
archivos = dir(fullfile(carpeta, '*.txt'));

% Celda donde se almacenarán los datos
datos = cell(length(archivos),1);

% Nombres de los archivos
nombres = strings(length(archivos),1);

% Leer cada archivo
for k = 1:length(archivos)

    nombreArchivo = fullfile(carpeta, archivos(k).name);

    datos{k} = readmatrix(nombreArchivo,...
                          'FileType','text',...
                          'CommentStyle','#');

    nombres(k) = archivos(k).name;

end

disp('Archivos cargados correctamente.')

%=====
%                PROCESAMIENTO DE SEÑALES
%=====

Fs = 1000; % Se define una frecuencia de muestreo de 1000 Hz

for k = 1:length(datos)

    fprintf('\nProcesando archivo %d de %d...\n',k,length(datos));
    fprintf('%s\n',nombres(k));

    M = datos{k};
    EMG = M(:,6);

    t = (0:length(EMG)-1)/Fs;

```

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%% =====
% PREPROCESAMIENTO INTERNO (SIN PLOTEO)
% =====

% Eliminación del offset
EMG0 = EMG - mean(EMG);

% Filtro Notch (Eliminación del ruido de red eléctrica)
wo = 60/(Fs/2);
bw = wo/35;

[b,a] = iirnotch(wo,bw);

EMG_notch = filtfilt(b,a,EMG0);

% Filtro pasabanda (Selección de la banda de interés para un EMG)
[b,a] = butter(4,[20 450]/(Fs/2), 'bandpass');

EMG_filtrada = filtfilt(b,a,EMG_notch);

%% =====
% 1. Señal original
% =====

figure;
plot(t,EMG)
grid on
xlabel('Tiempo (s)')
ylabel('Amplitud')
title(['Señal original - ' char(nombres(k))])

saveas(gcf,fullfile(carpeta,...
    erase(nombres(k),".txt")+ "_01_Original.png"));

%% =====
% 2. Envolvente
% =====

EMG_rect = abs(EMG_filtrada);

ventana = round(0.05*Fs);

EMG_env = movmean(EMG_rect,ventana);

figure;
plot(t,EMG_rect)
hold on
plot(t,EMG_env, 'LineWidth',2)
grid on
legend('Rectificada','Media móvil')

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xlabel('Tiempo (s)')
ylabel('Amplitud')
title(['Envolvente - ' char(nombres(k))])

saveas(gcf,fullfile(carpeta,...
    erase(nombres(k),".txt")+ "_02_Envolvente.png"));

%% =====
% 3. FFT
% =====

N = length(EMG_filtrada);

Y = fft(EMG_filtrada);

f = (0:N-1)*(Fs/N);

P = abs(Y)/N;

figure;
plot(f(1:N/2),P(1:N/2))
grid on
xlabel('Frecuencia (Hz)')
ylabel('|X(f)|')
title(['FFT - ' char(nombres(k))])
xlim([0 500])

saveas(gcf,fullfile(carpeta,...
    erase(nombres(k),".txt")+ "_03_FFT.png"));

%% =====
% 4. Periodograma
% =====

figure;
periodogram(EMG_filtrada,[],[],Fs)

title(['Periodograma - ' char(nombres(k))])

saveas(gcf,fullfile(carpeta,...
    erase(nombres(k),".txt")+ "_04_Periodograma.png"));

%% =====
% 5. Welch
% =====

figure;
pwelch(EMG_filtrada,[],[],[],Fs)

title(['PSD mediante Welch - ' char(nombres(k))])

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```

saveas(gcf,fullfile(carpeta,...
    erase(nombres(k),".txt")+ "_05_Welch.png"));

%% =====
% 6. Espectrograma
% =====

figure;
spectrogram(EMG_filtrada,...
    hamming(256),...
    200,...
    512,...
    Fs,...
    'yaxis')

title(['Espectrograma - ' char(nombres(k))])

saveas(gcf,fullfile(carpeta,...
    erase(nombres(k),".txt")+ "_06_Espectrograma.png"));

%% =====
% 7. Comparación (Modificación de la señal Original para análisis)
% =====

figure

subplot(2,1,1)
plot(t,EMG)
grid on
title('Señal original')

subplot(2,1,2)
plot(t,EMG_filtrada)
grid on
title('Señal filtrada')
xlabel('Tiempo (s)')

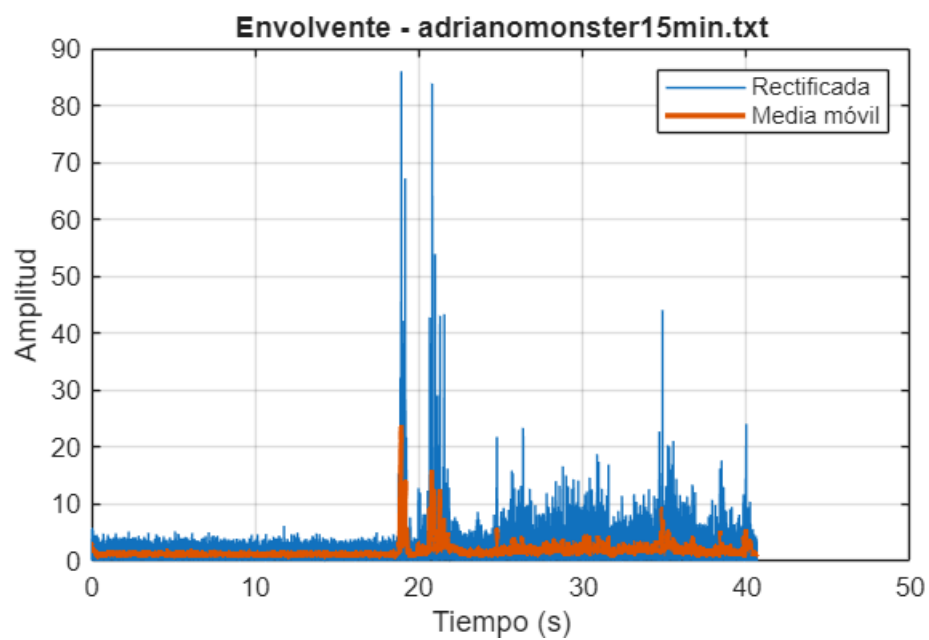
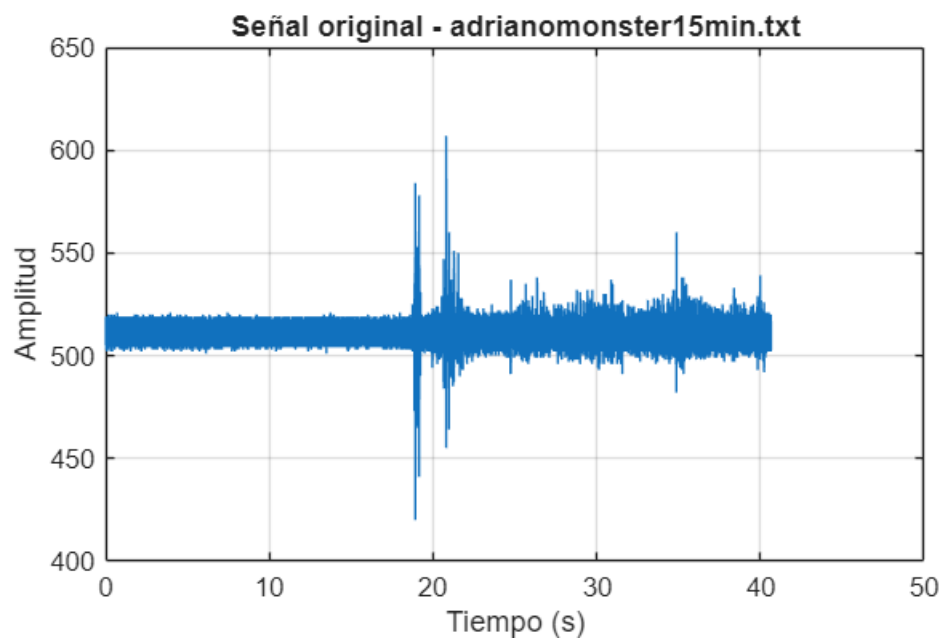
sgtitle(char(nombres(k)))

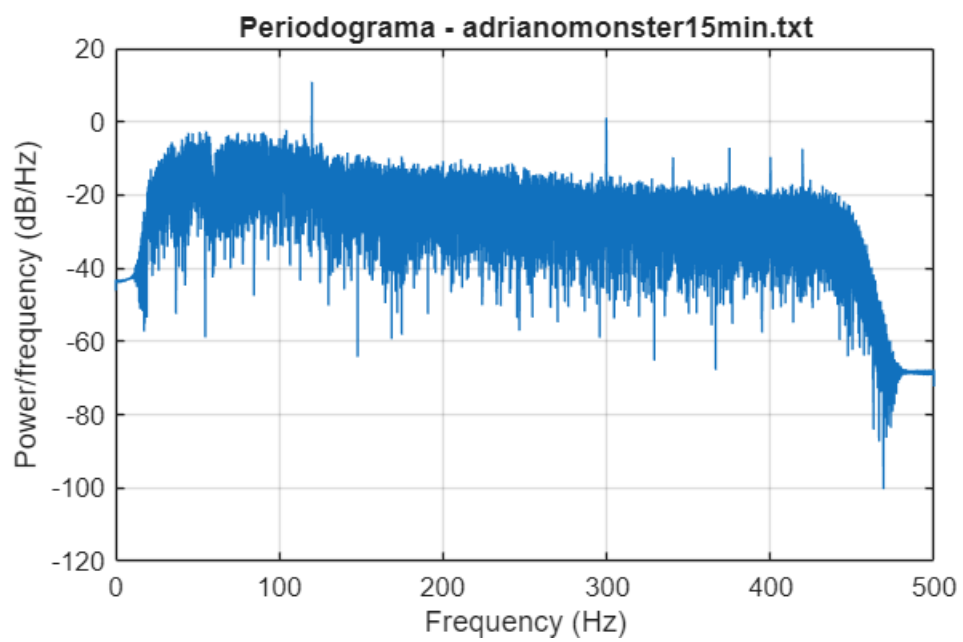
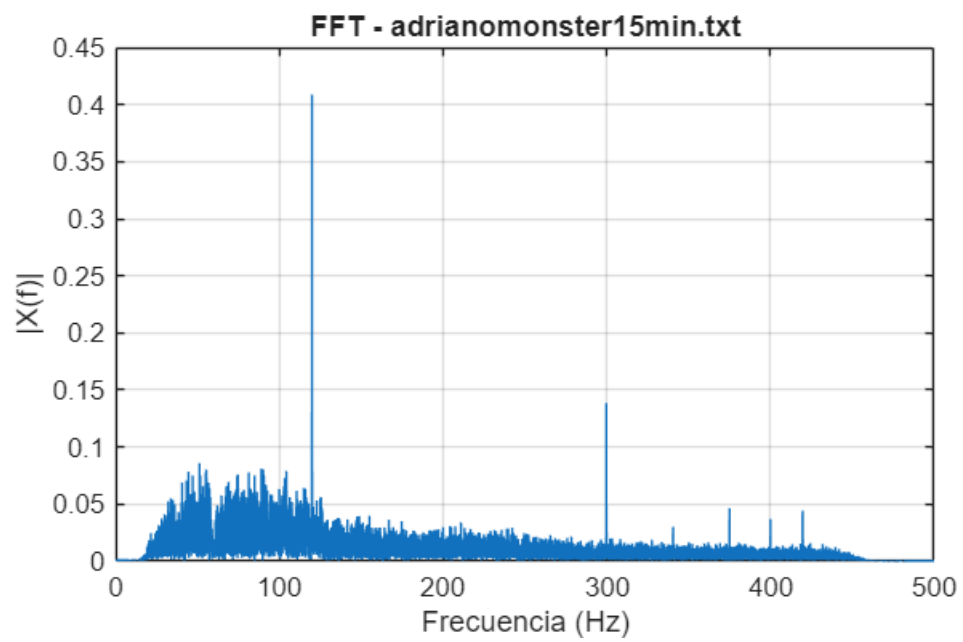
saveas(gcf,fullfile(carpeta,...
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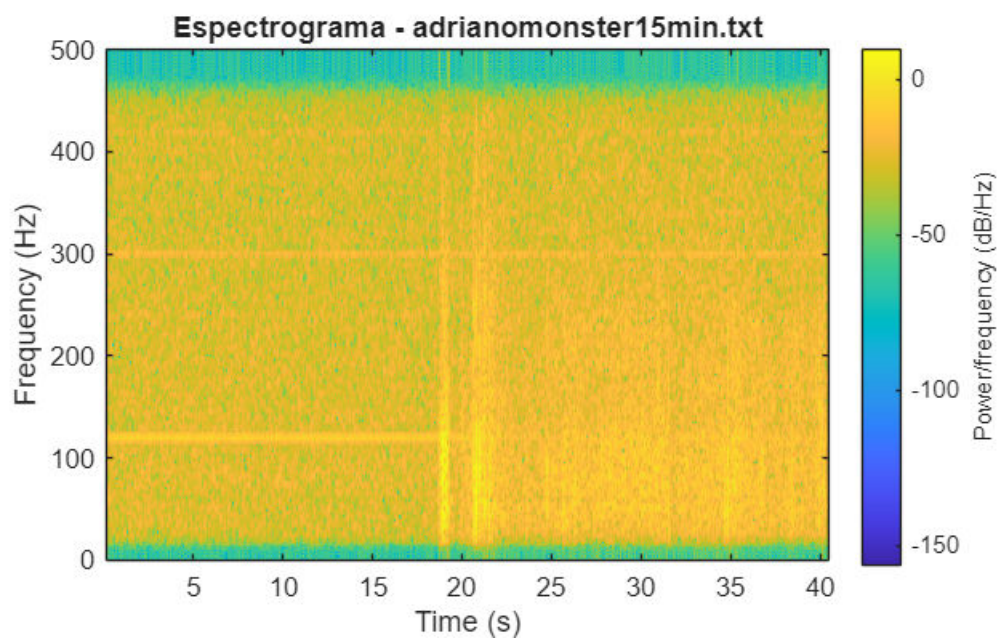
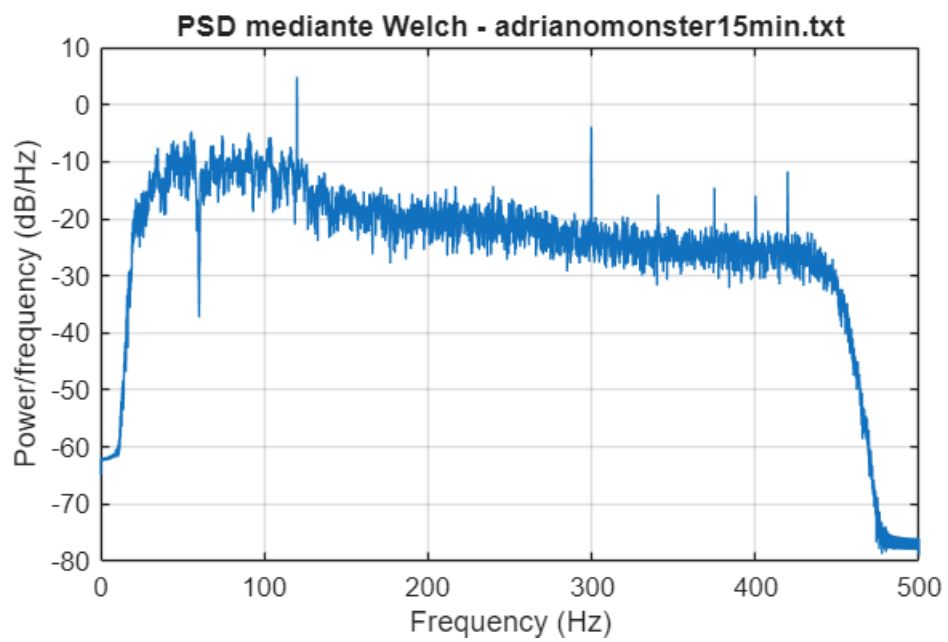
end

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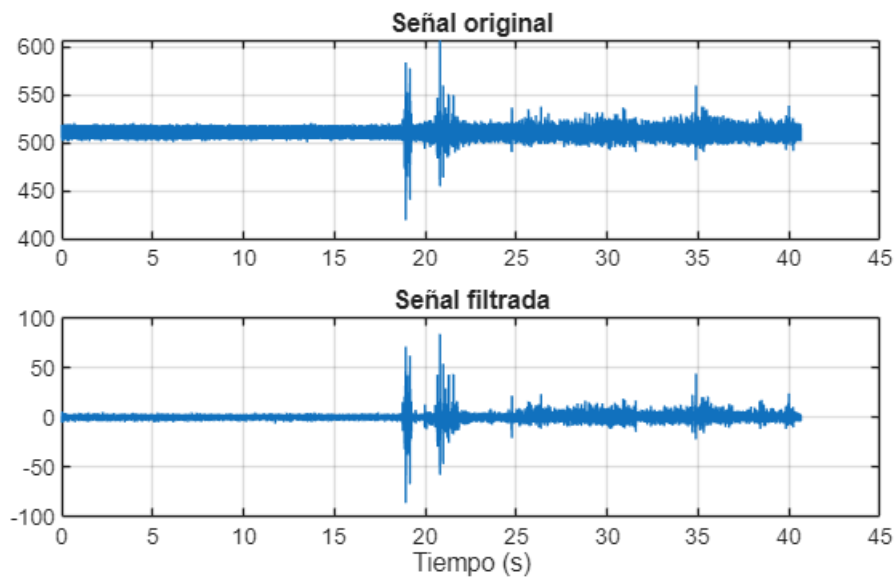
Procesando archivo 1 de 15...
adrianomonster15min.txt





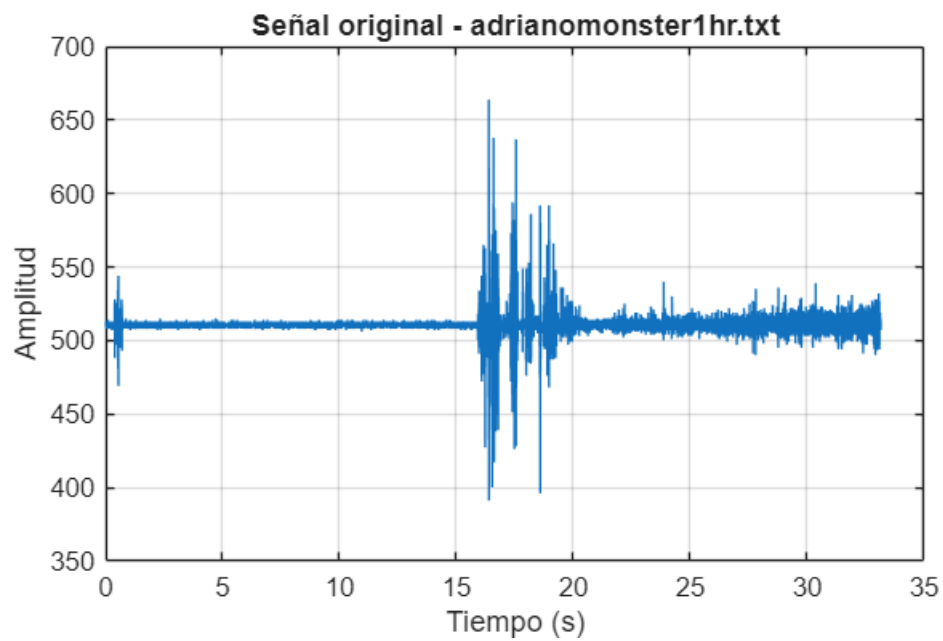


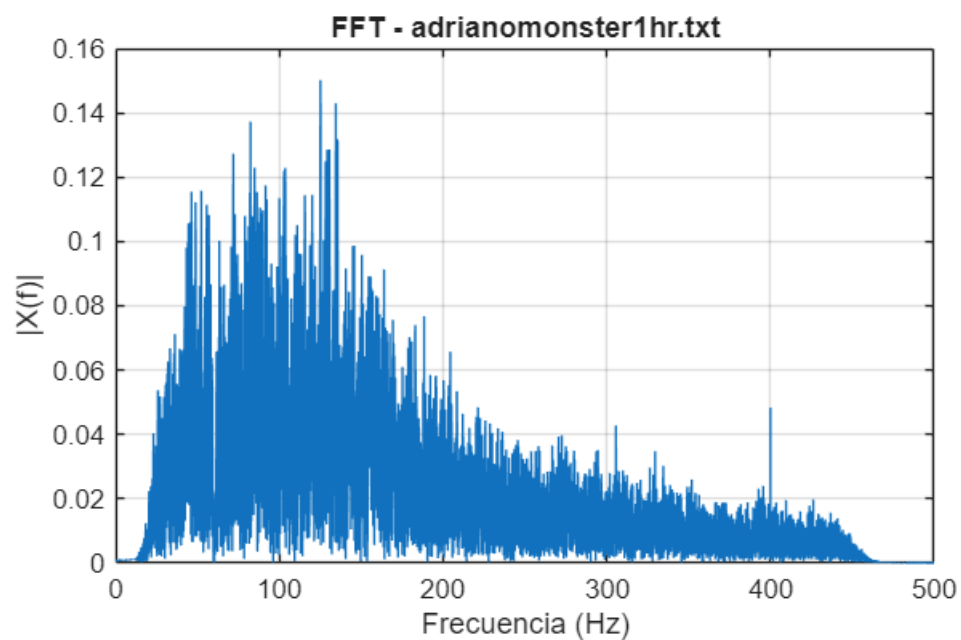
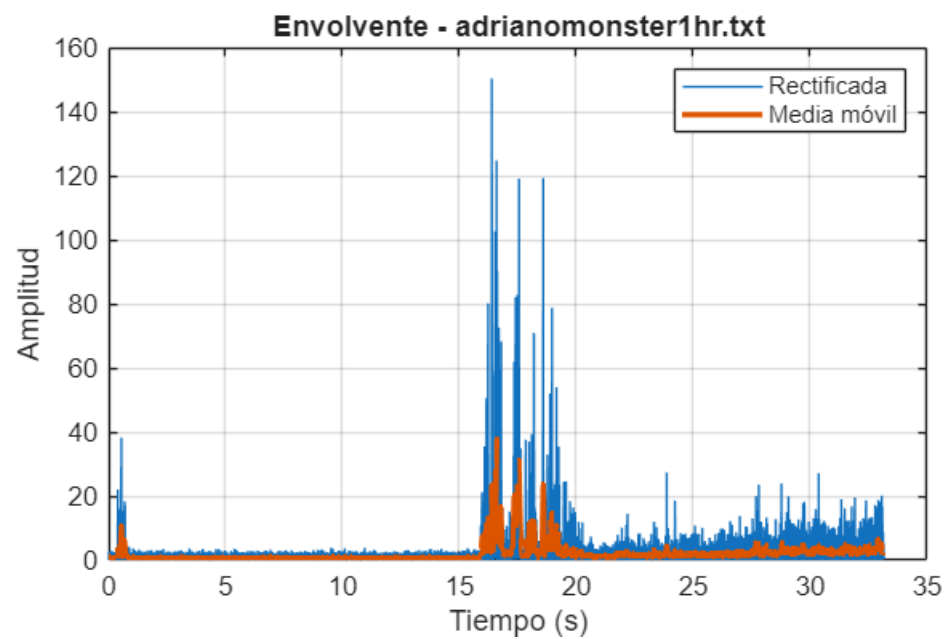
adrianomster15min.txt

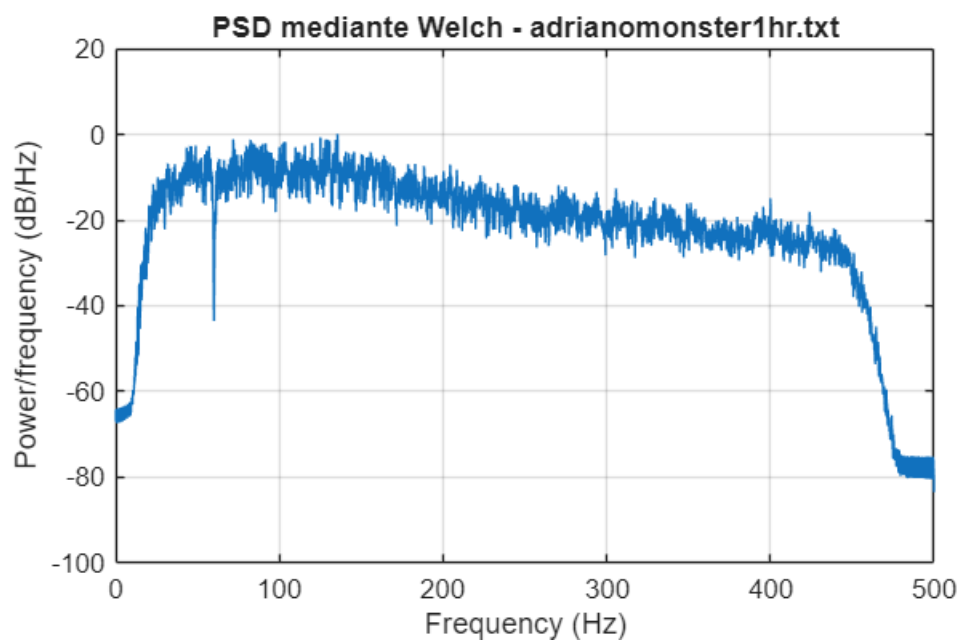
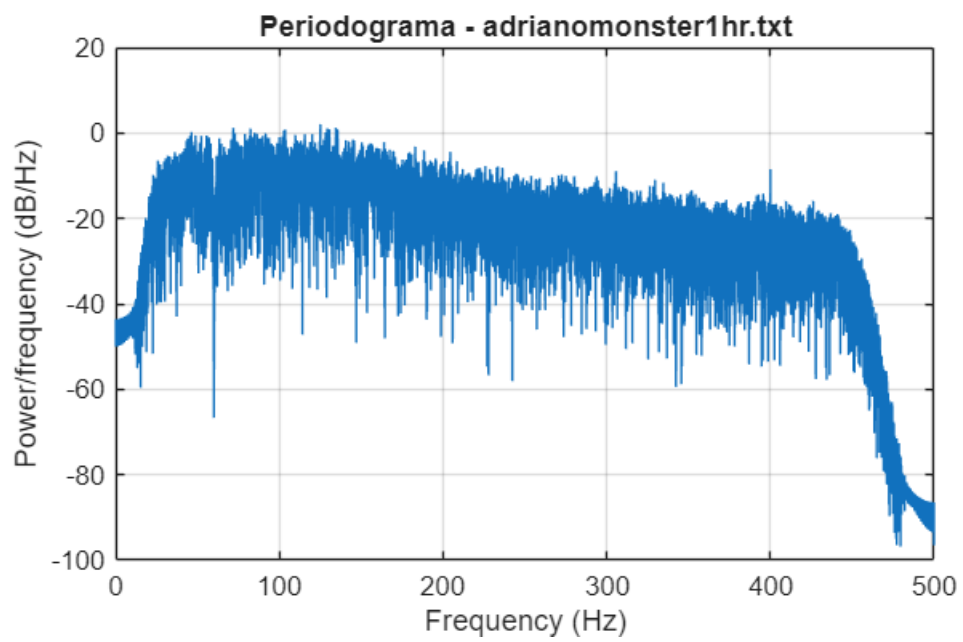


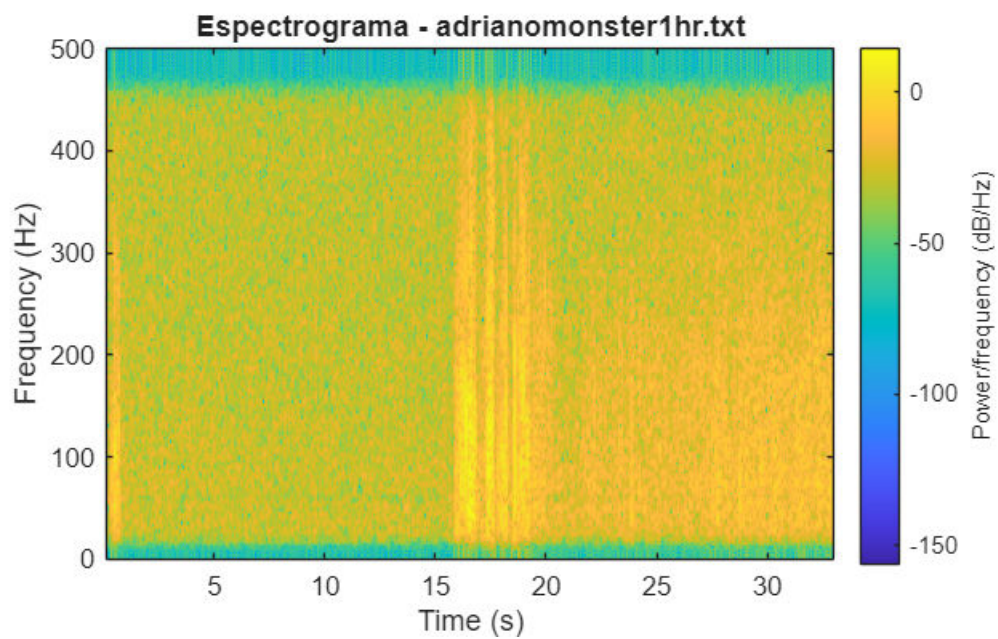
Procesando archivo 2 de 15...

adrianomster1hr.txt

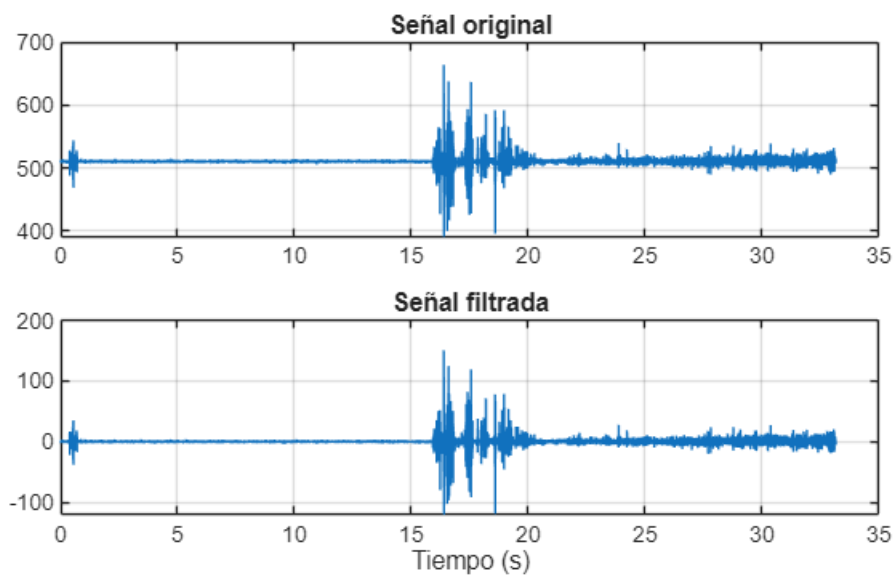




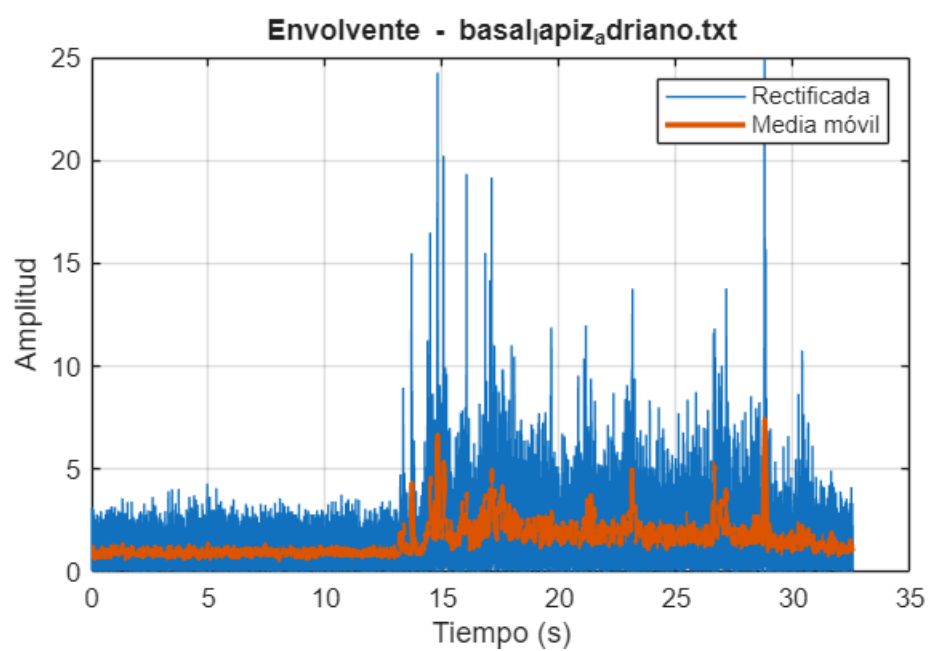
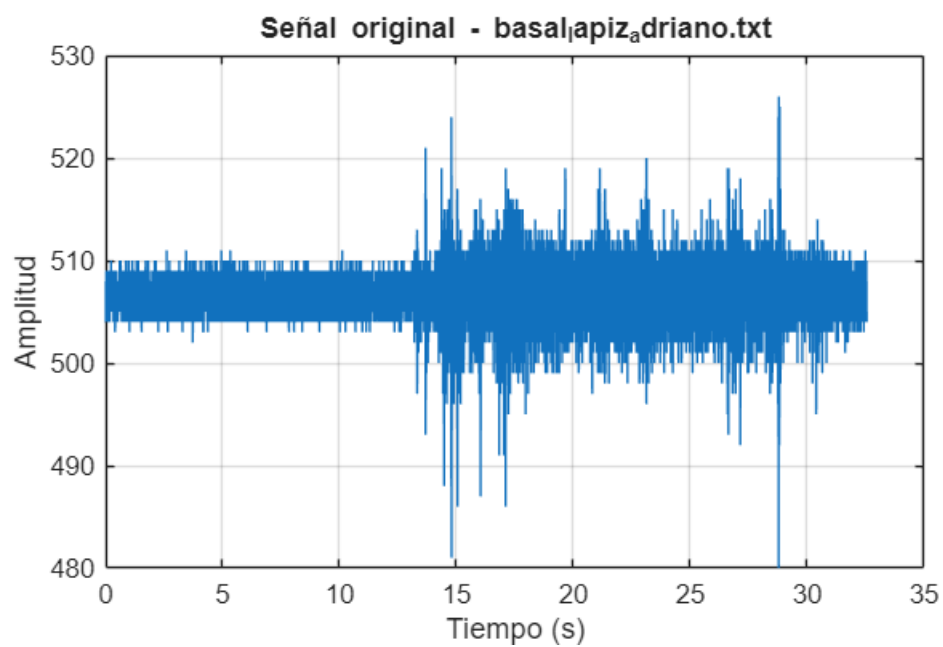


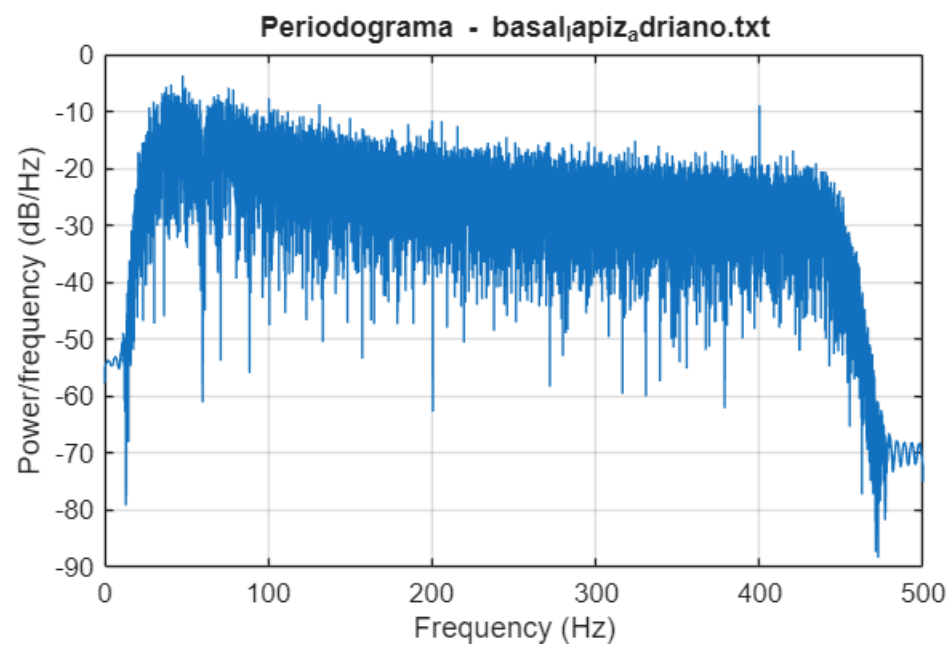
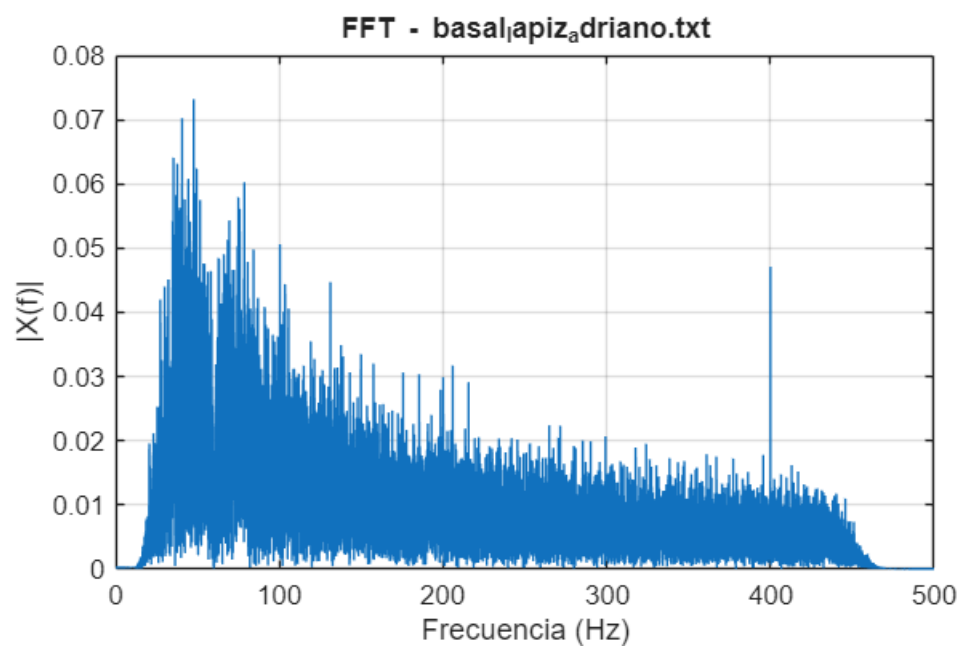


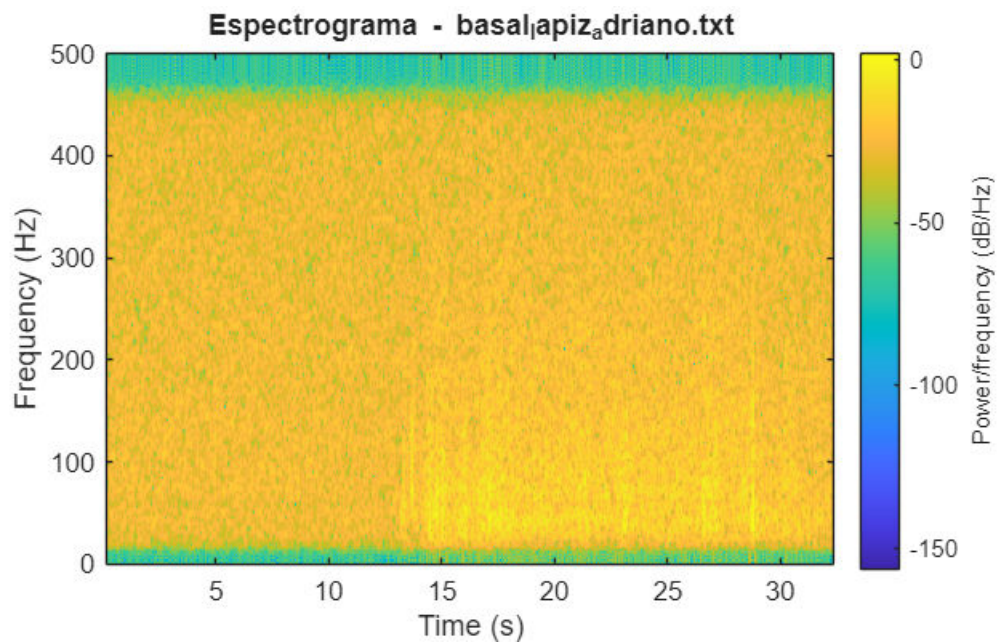
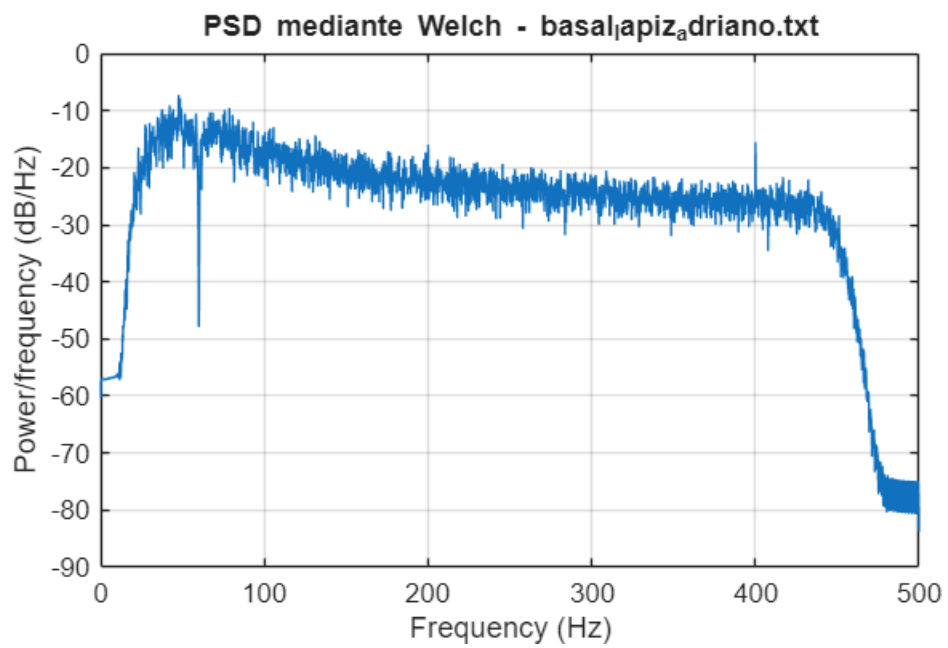
adrianommonster1hr.txt



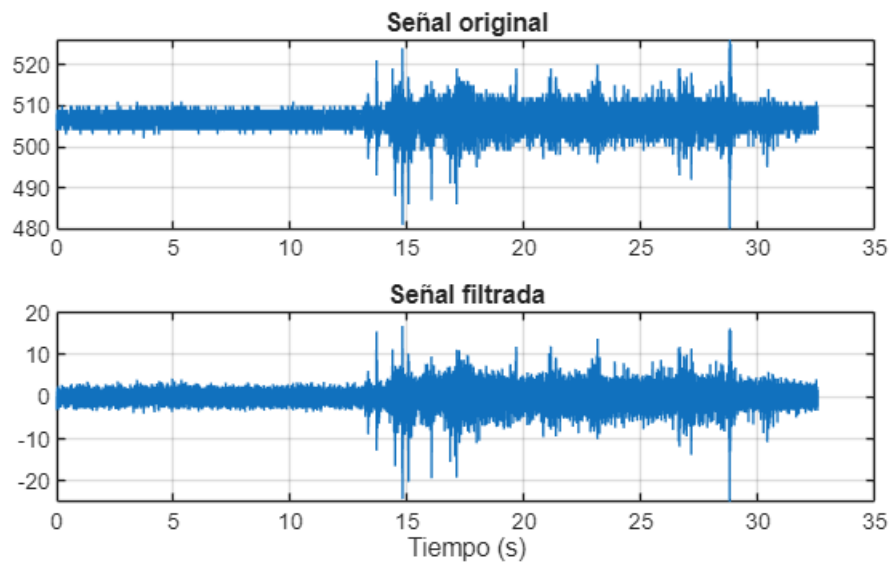
Procesando archivo 3 de 15...
basal_lapiz_adriano.txt





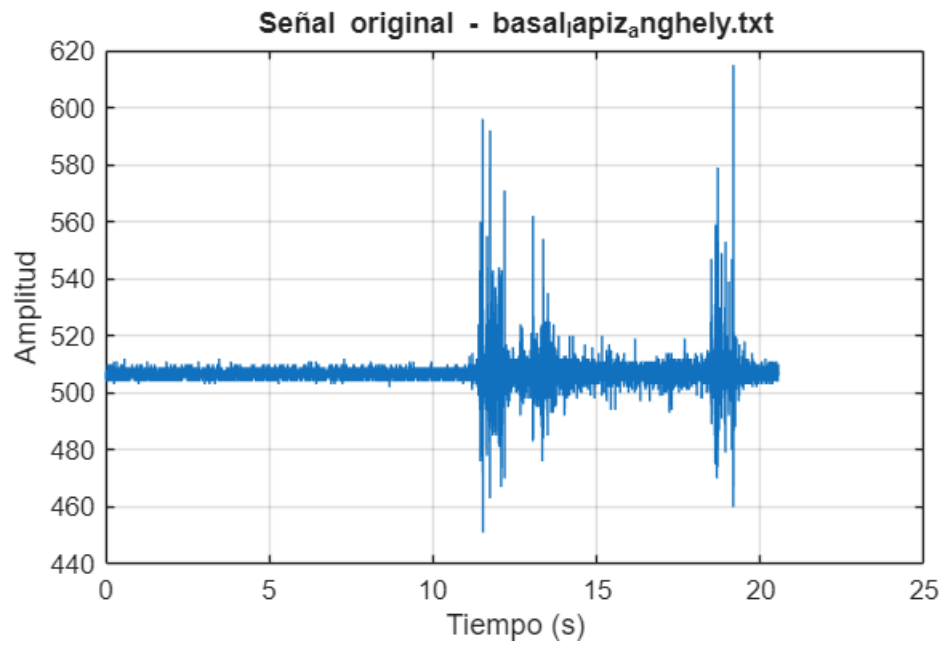


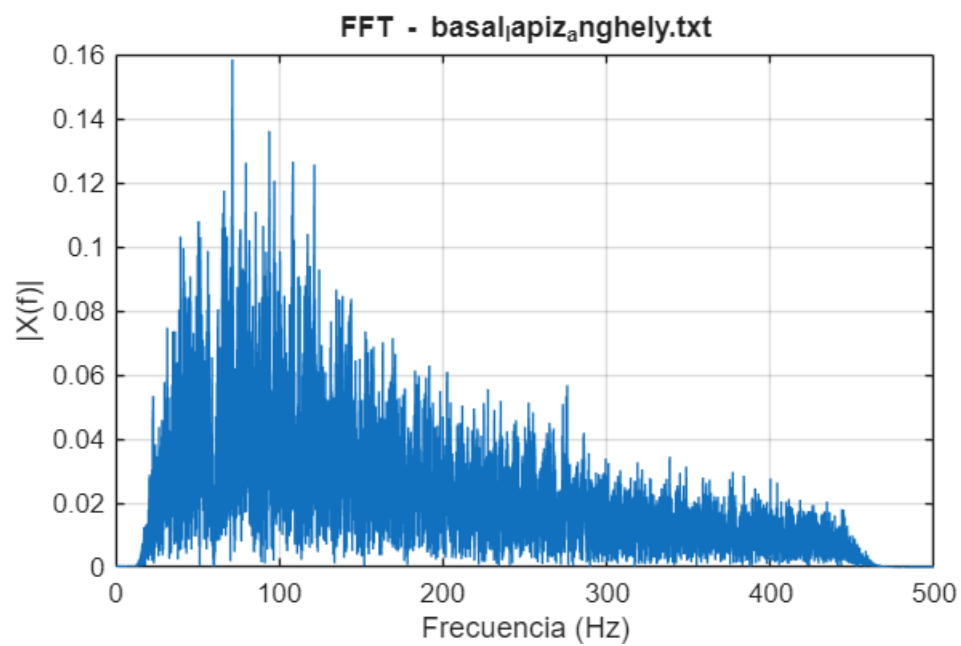
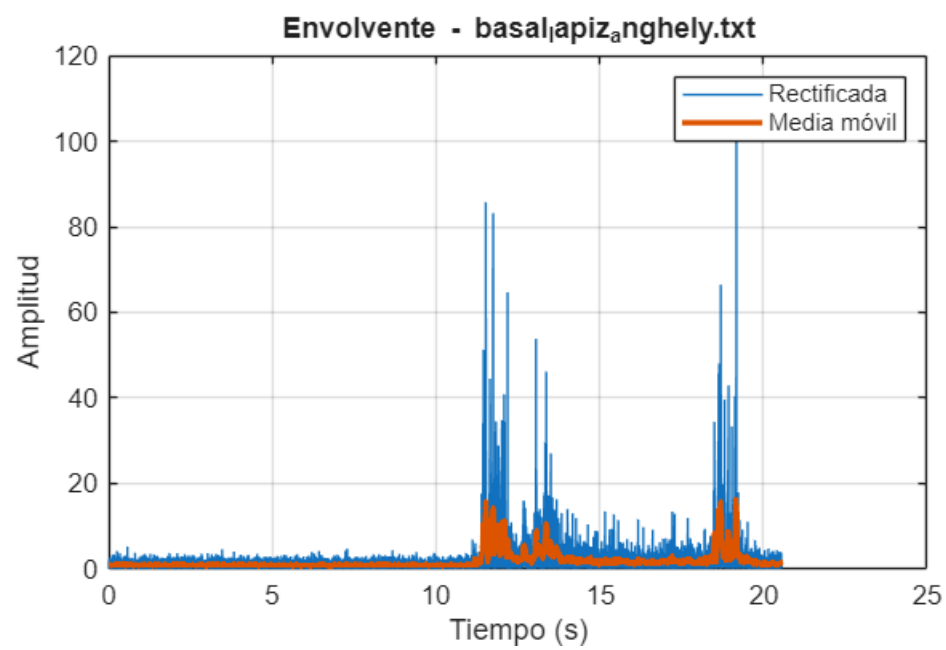
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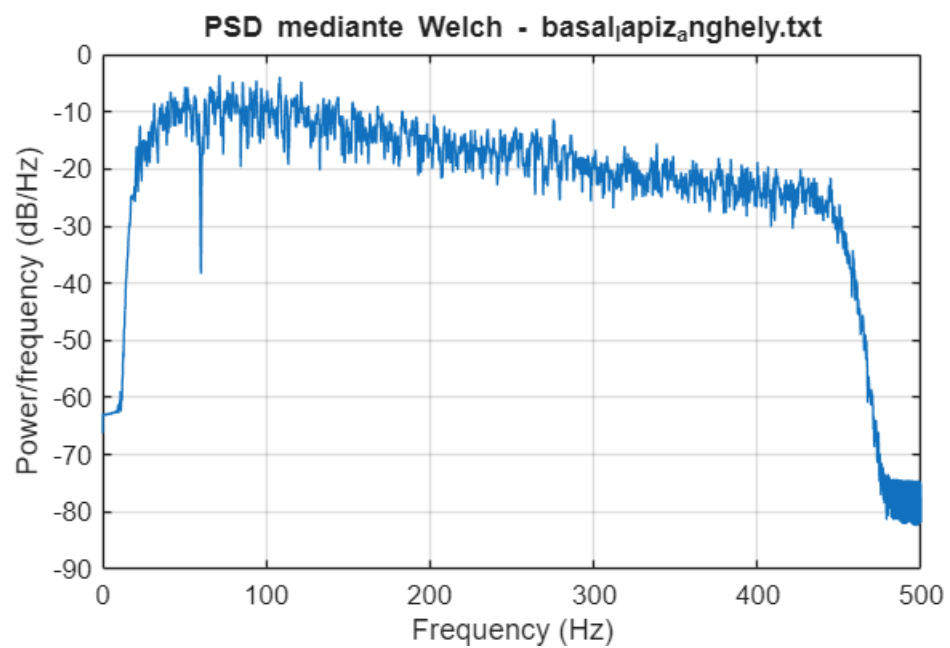
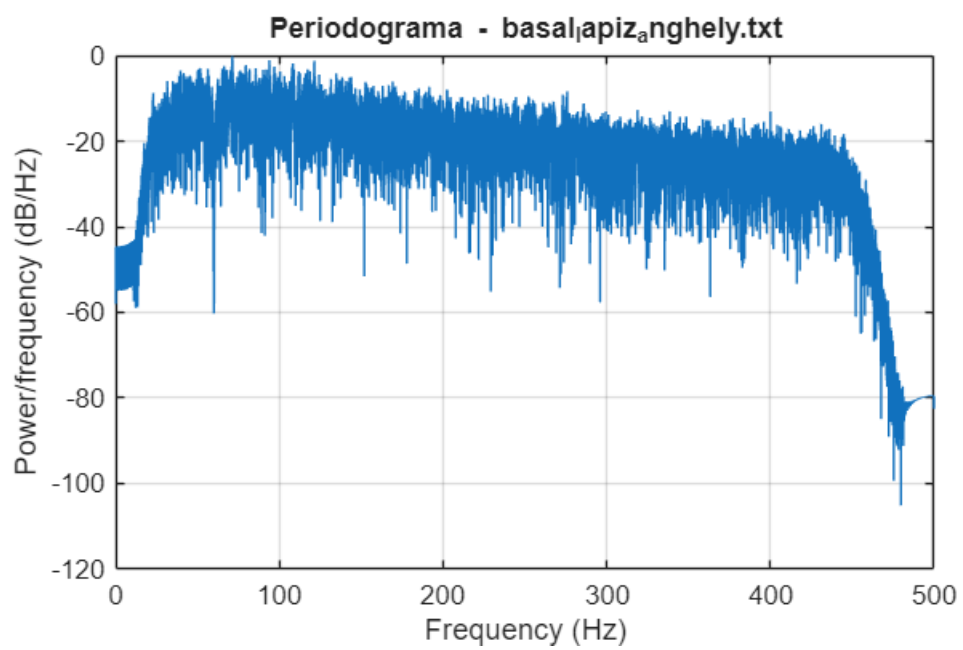


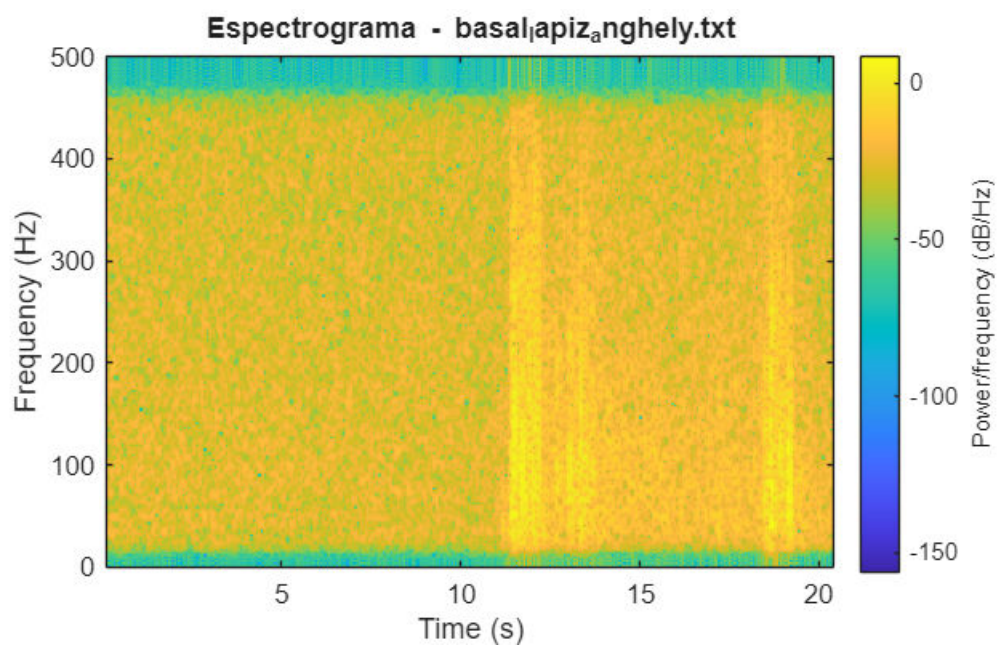
Procesando archivo 4 de 15...

basal_lapiz_anghely.txt

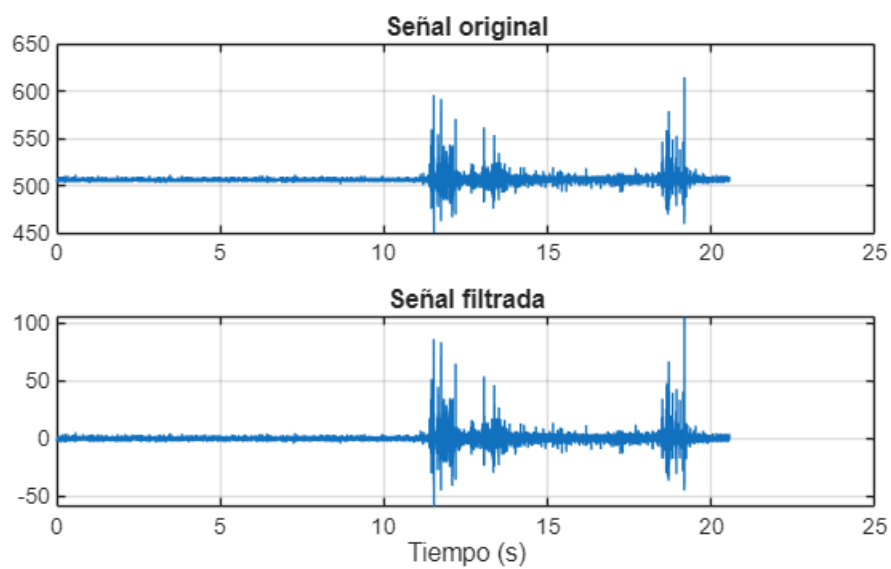




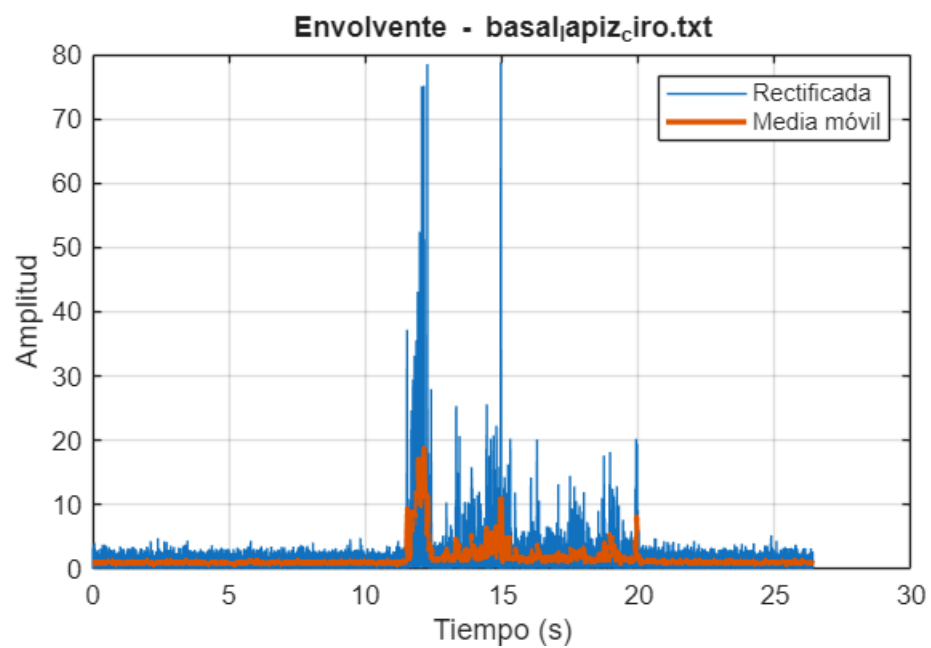
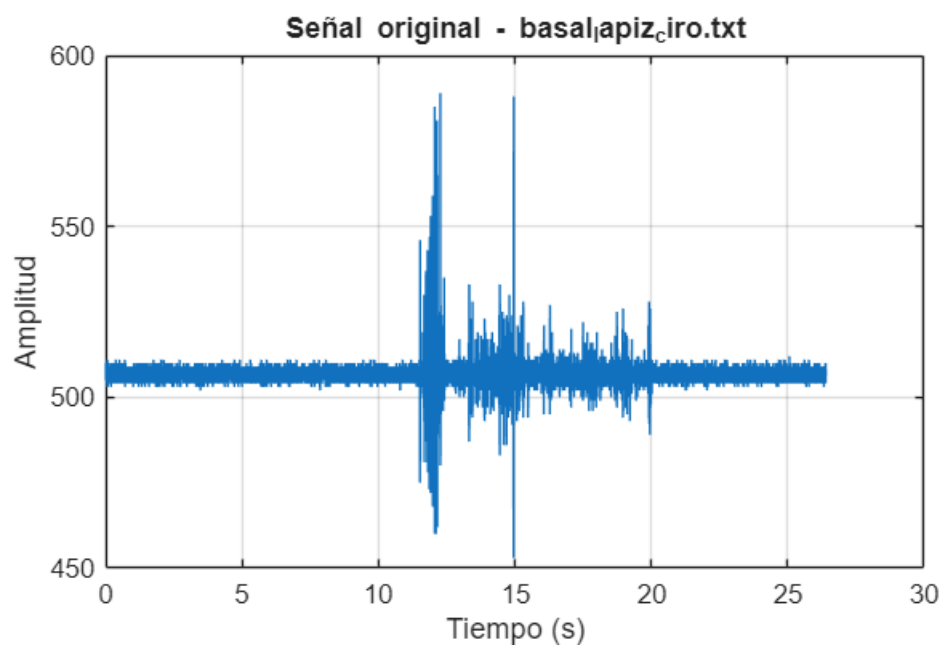


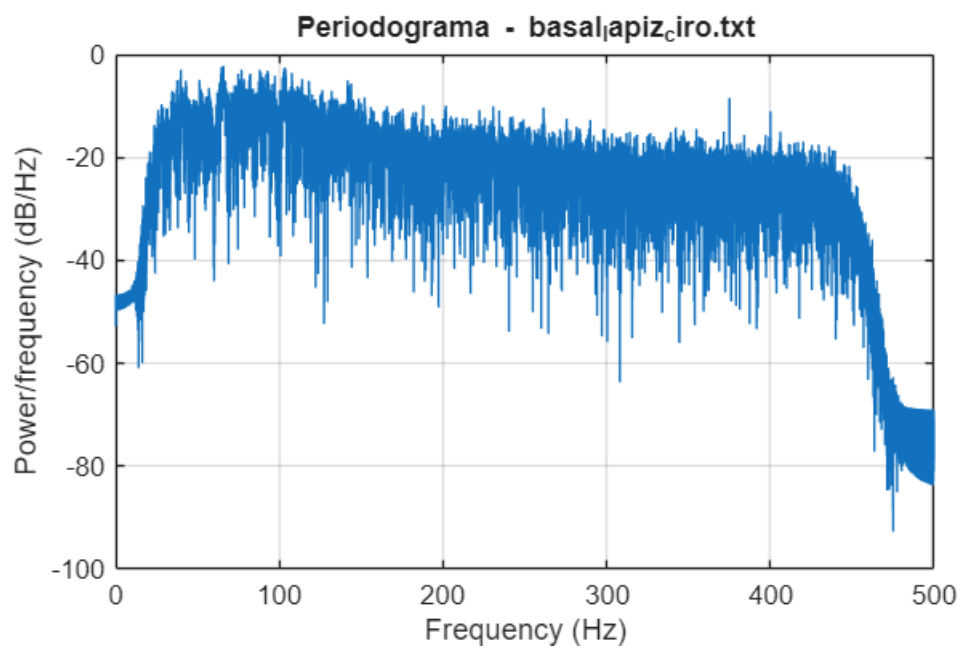
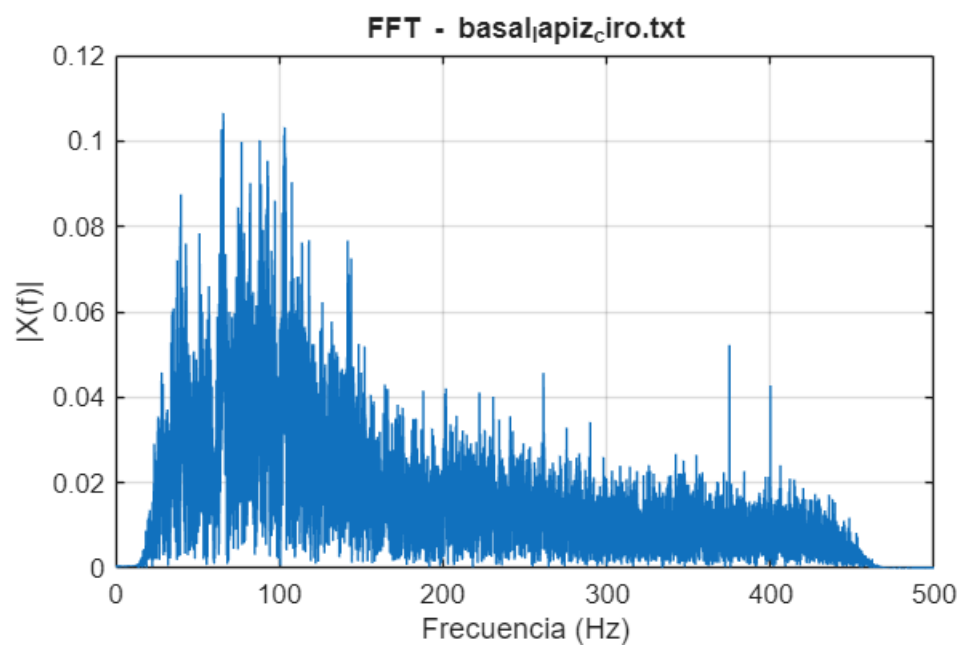


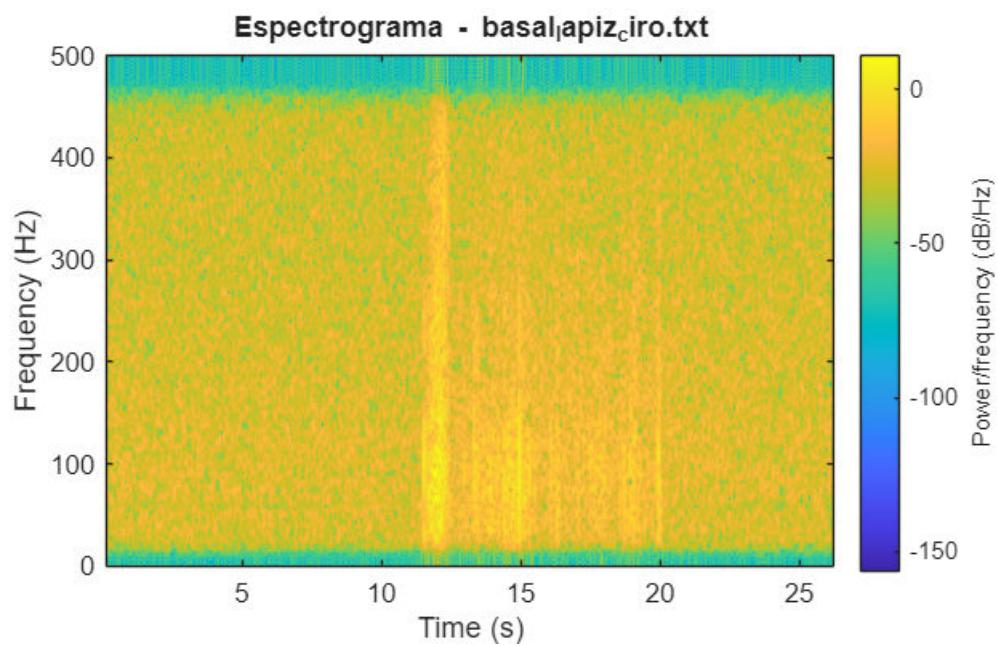
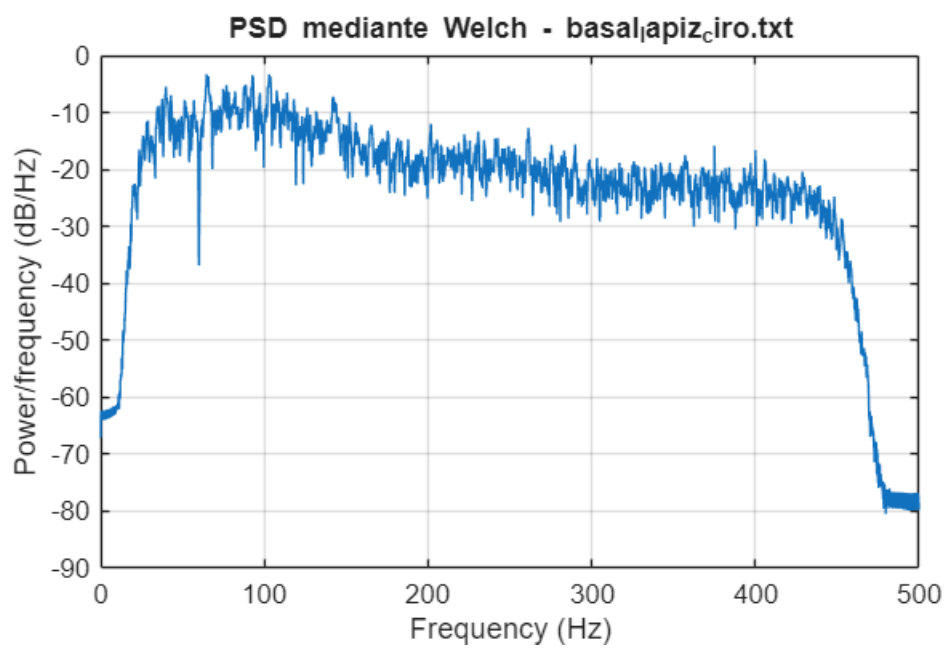
basal_lapiz_a_nghely.txt



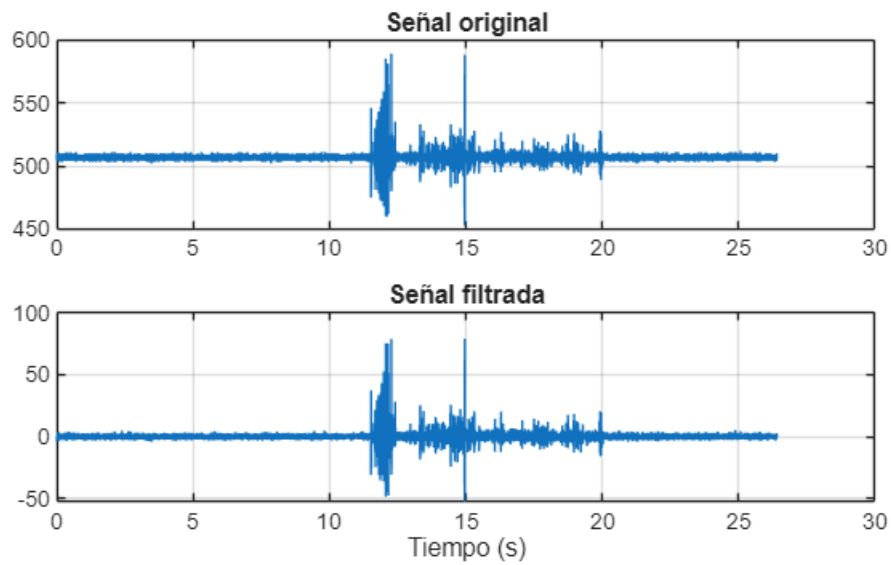
Procesando archivo 5 de 15...
basal_lapiz_ciro.txt



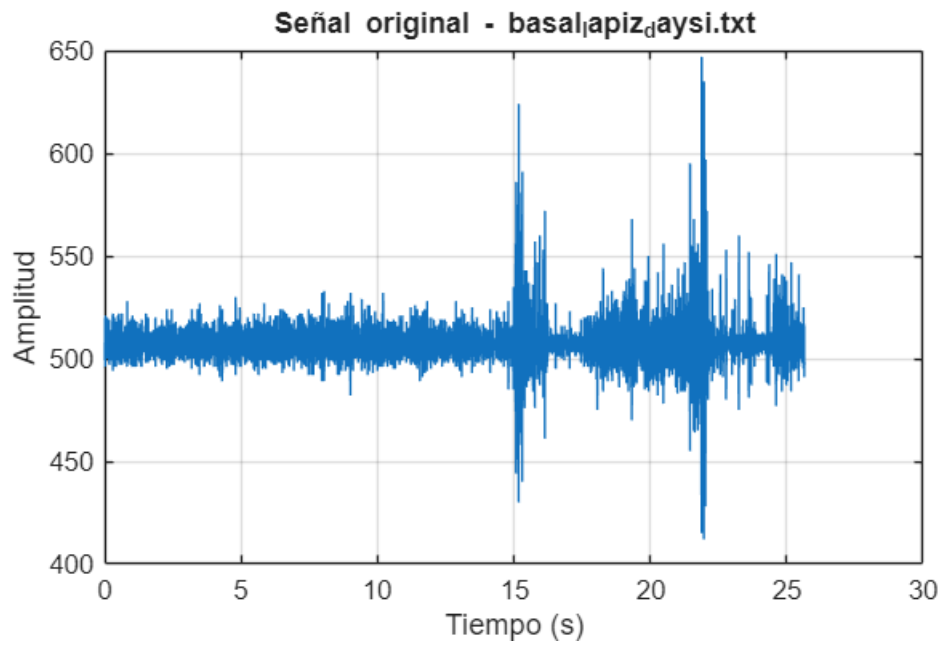


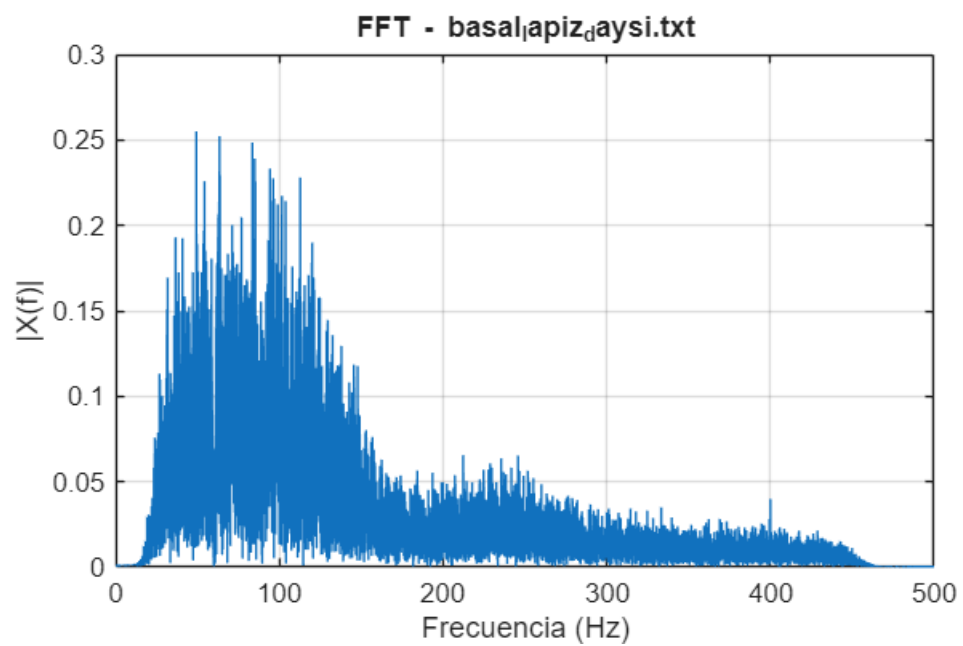
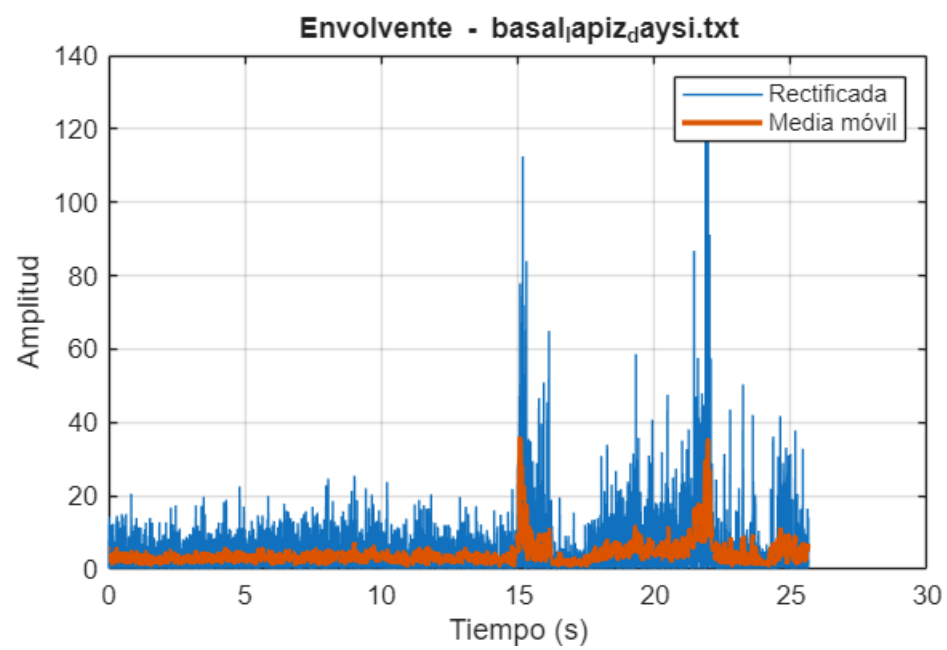


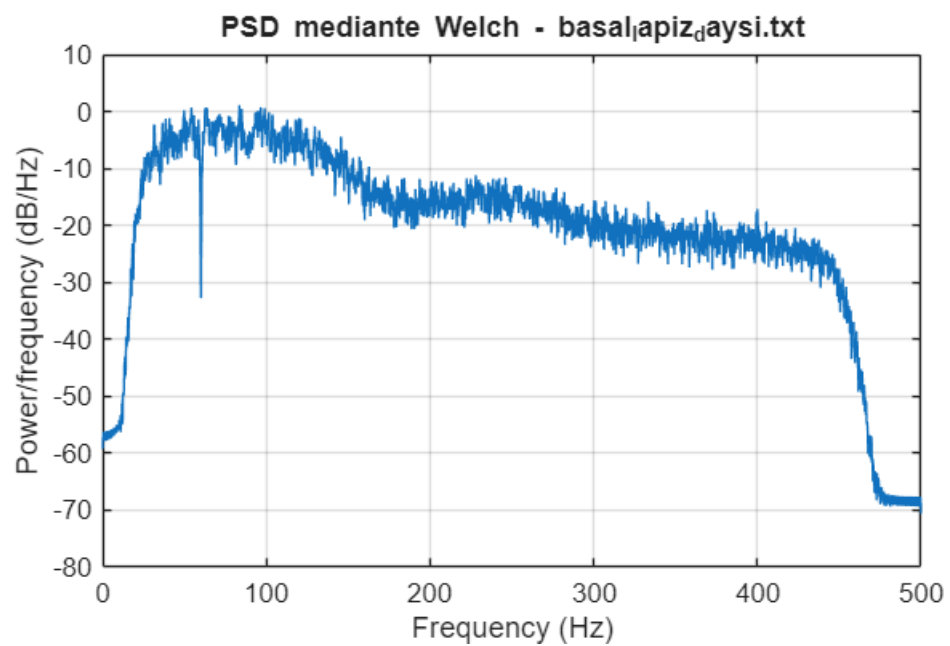
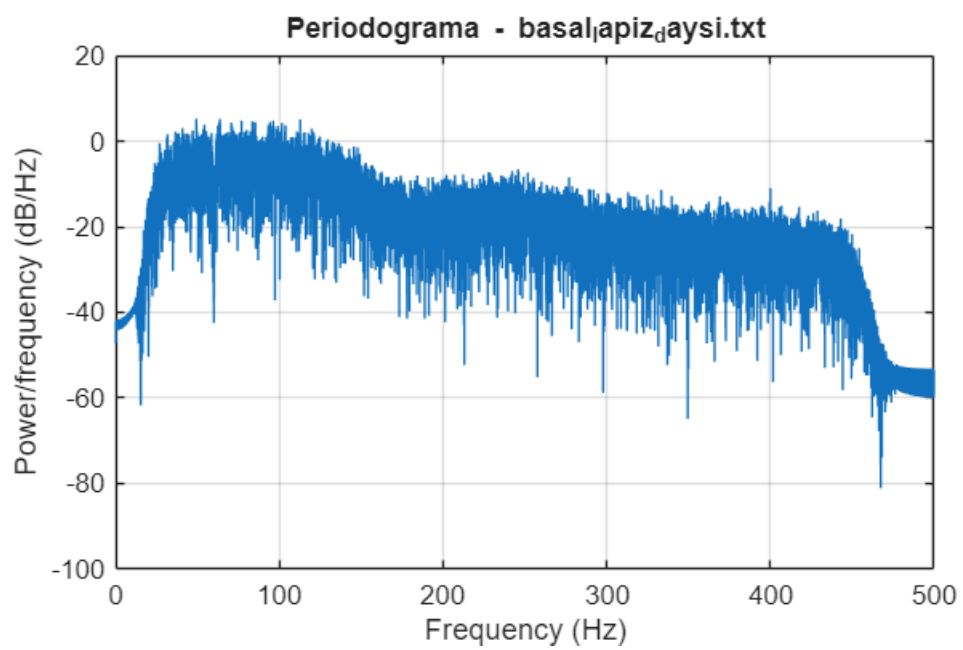
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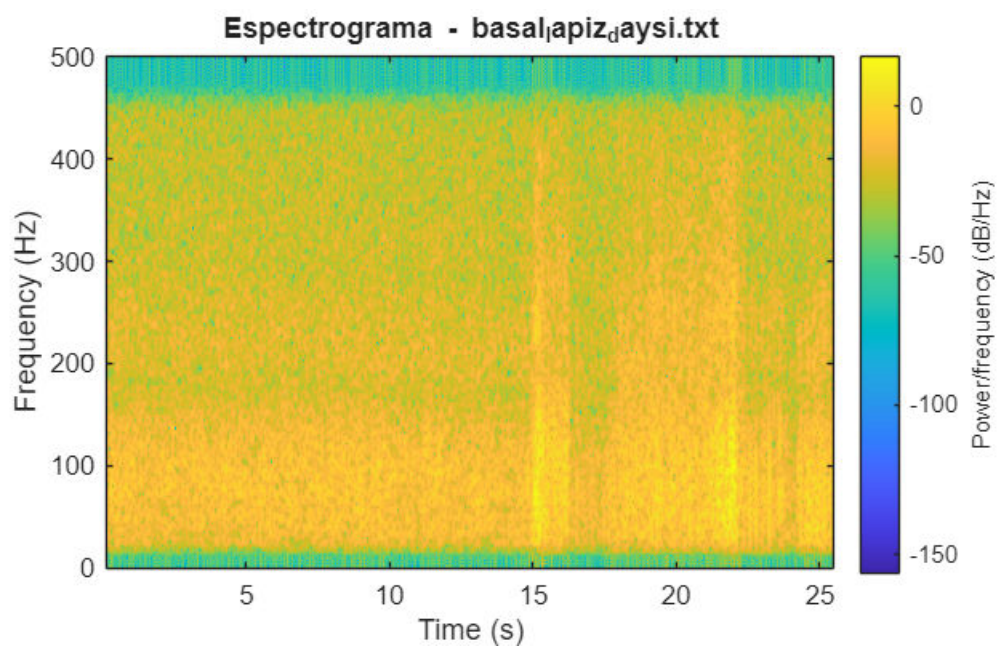


Procesando archivo 6 de 15...
basal_lapiz_daysi.txt

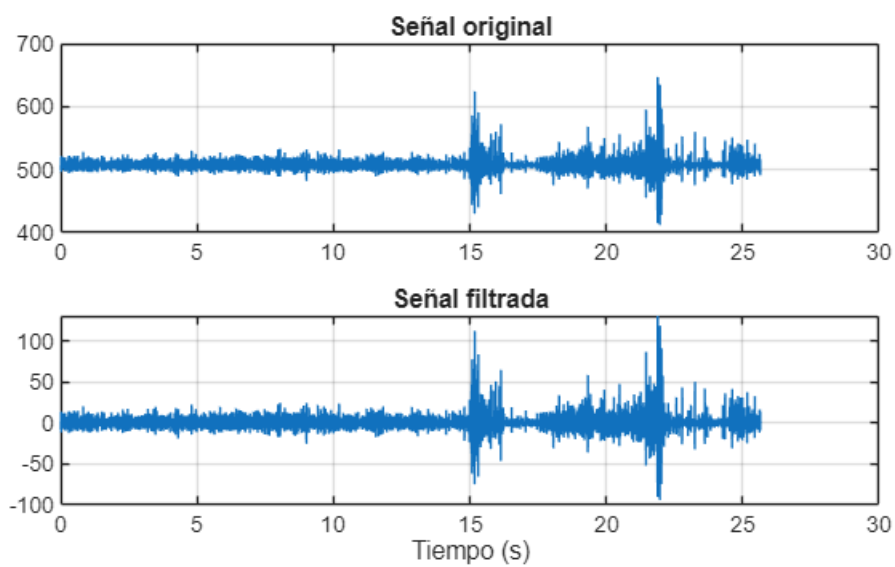




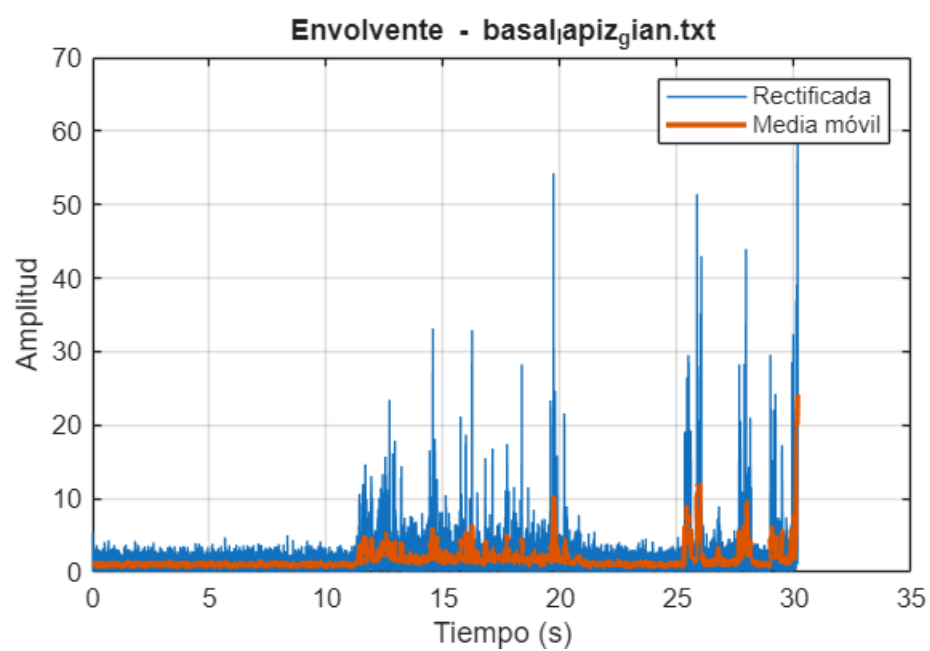
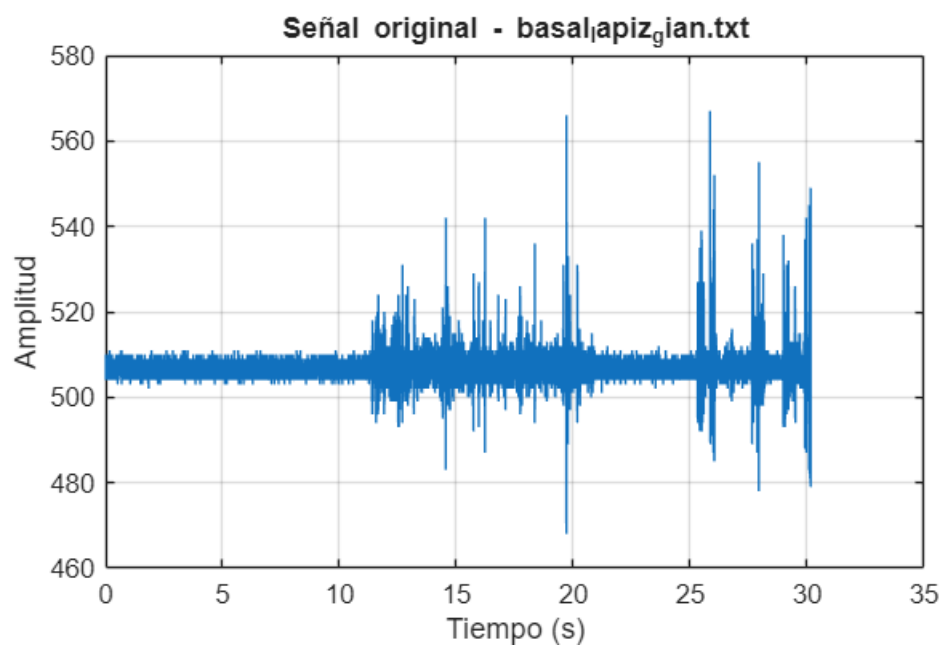


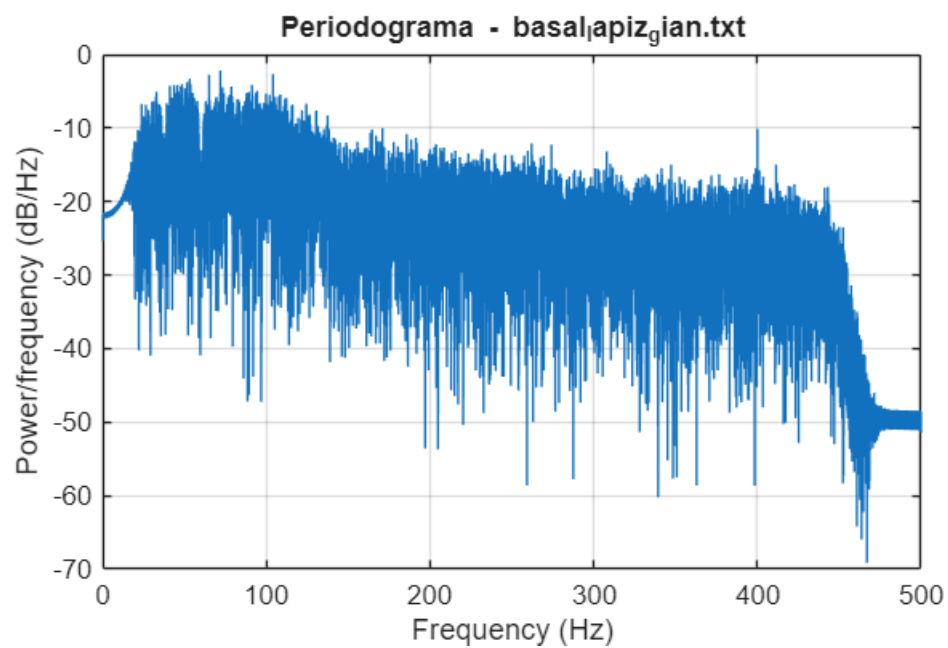
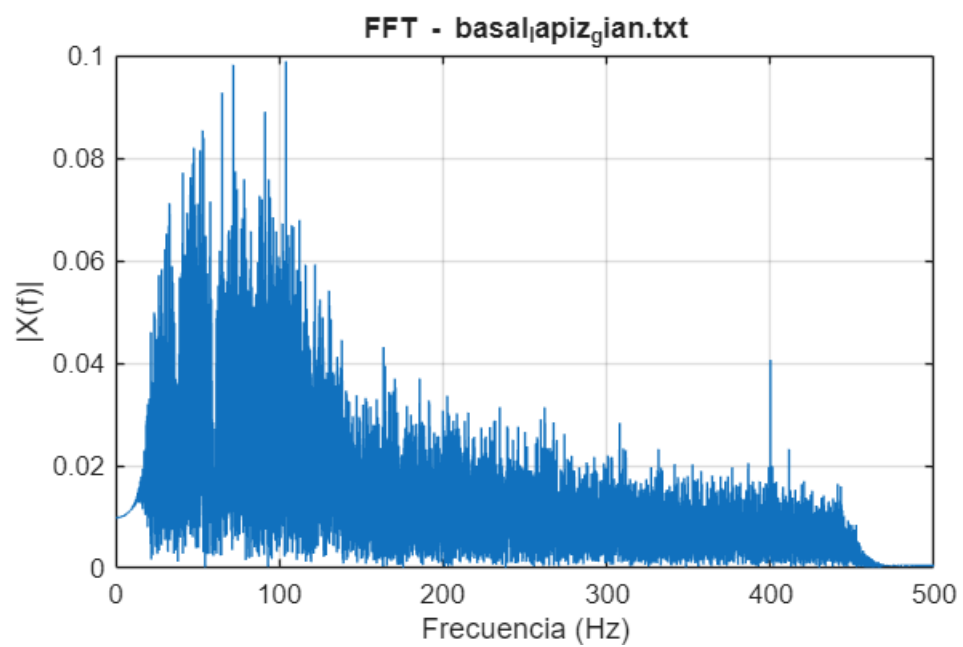


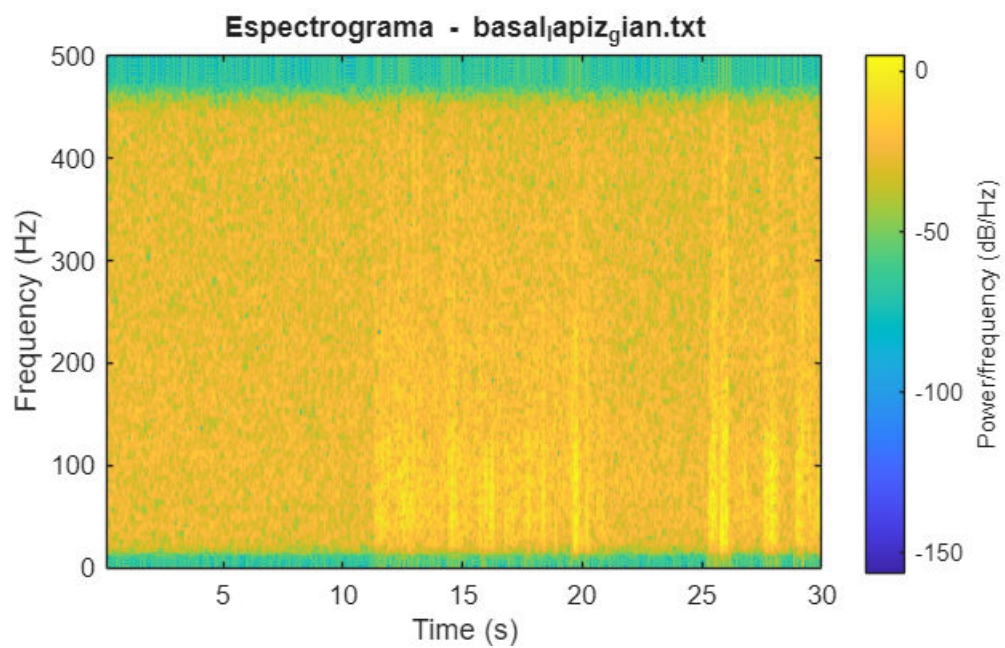
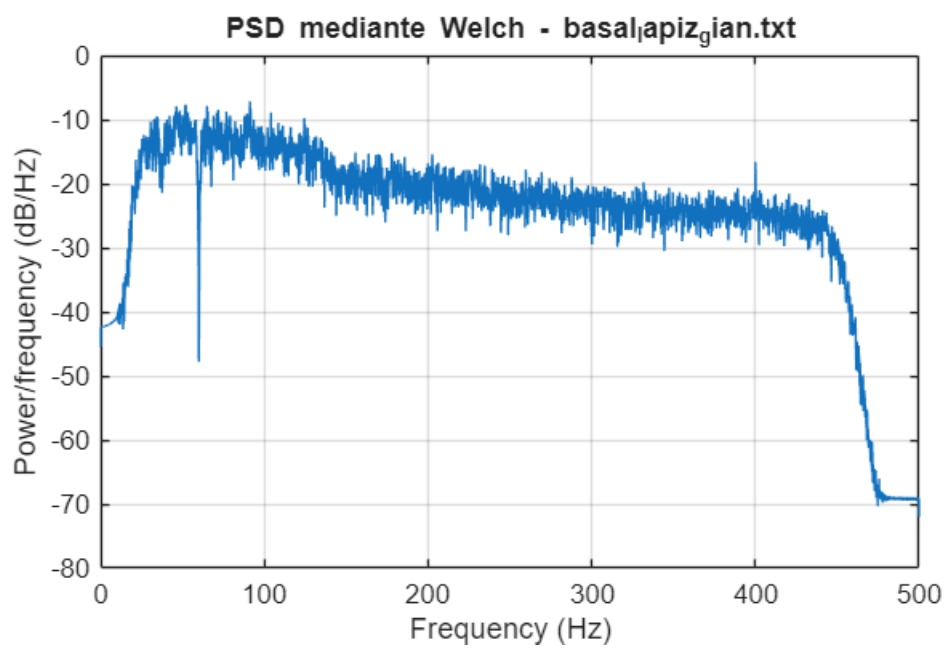
basal_lapiz_daysi.txt



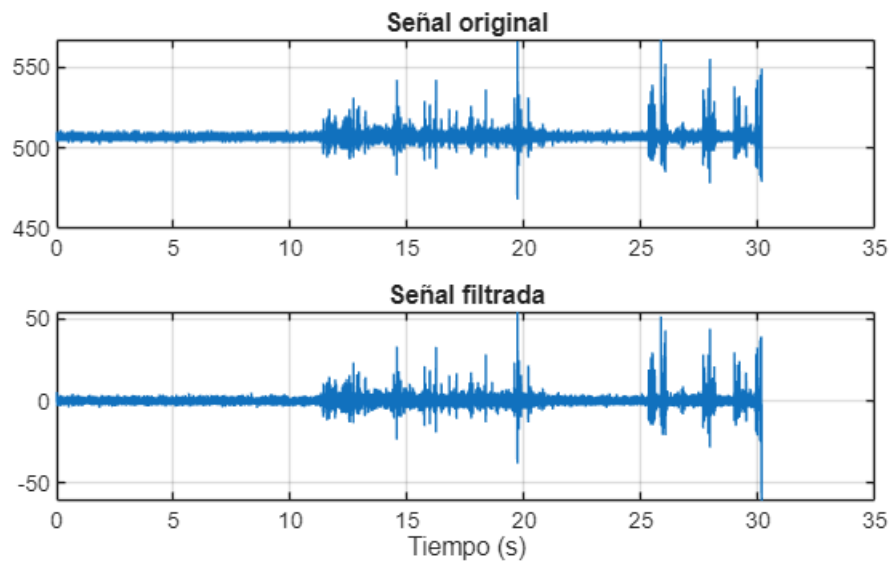
Procesando archivo 7 de 15...
basal_lapiz_gian.txt



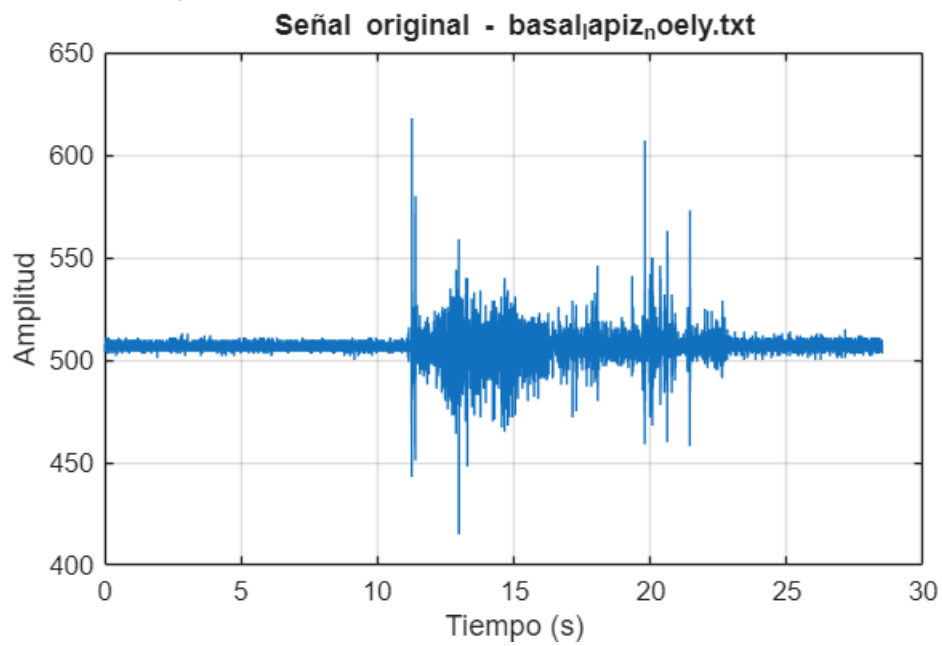


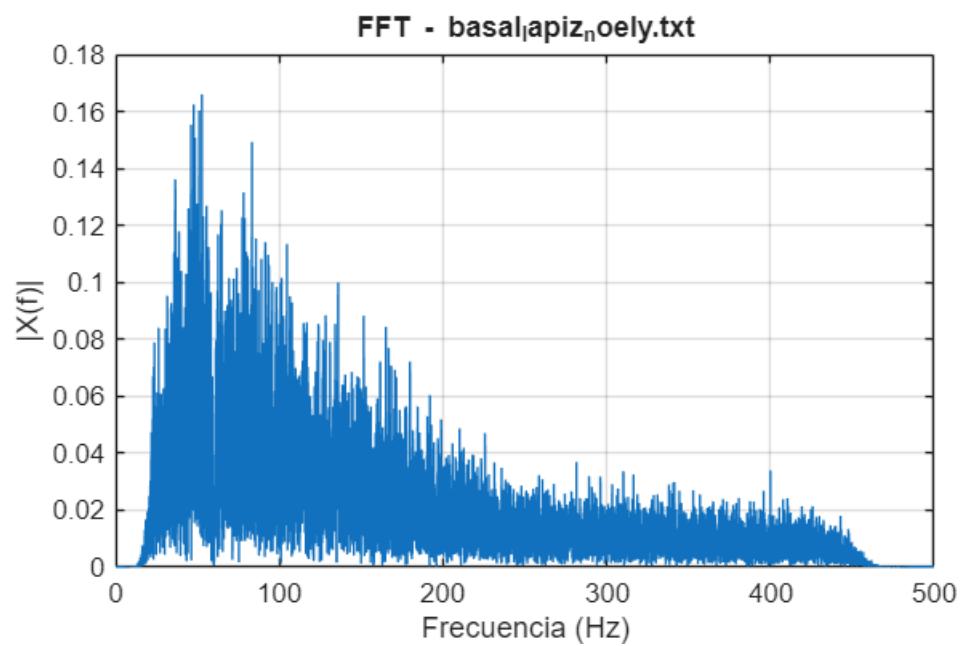
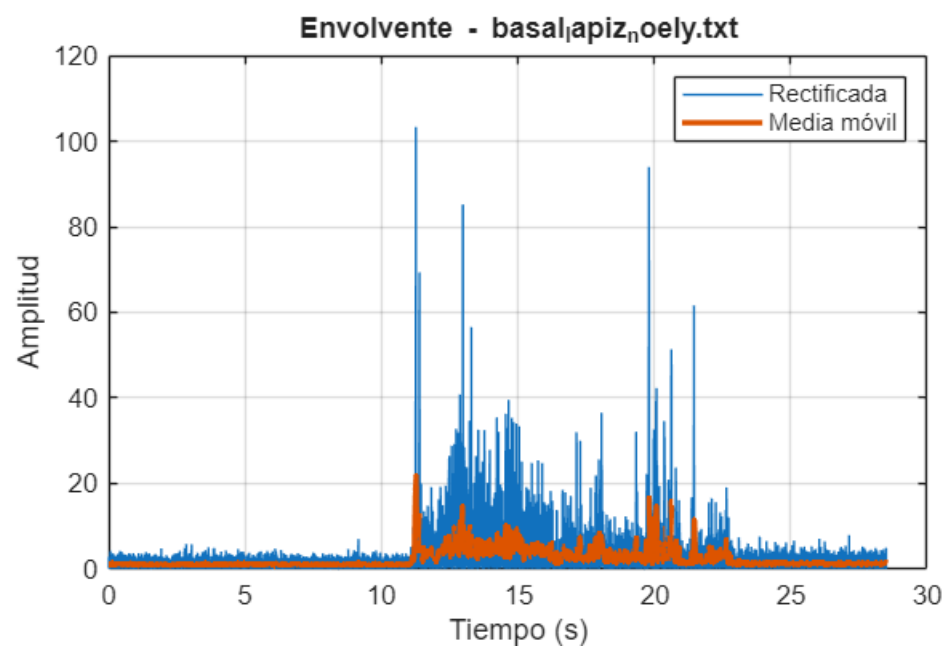


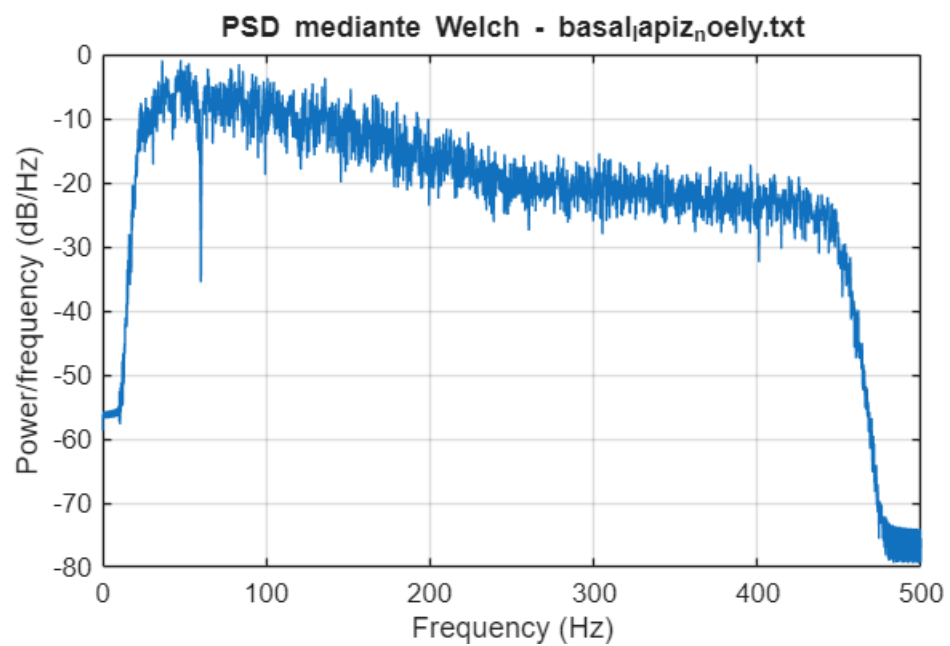
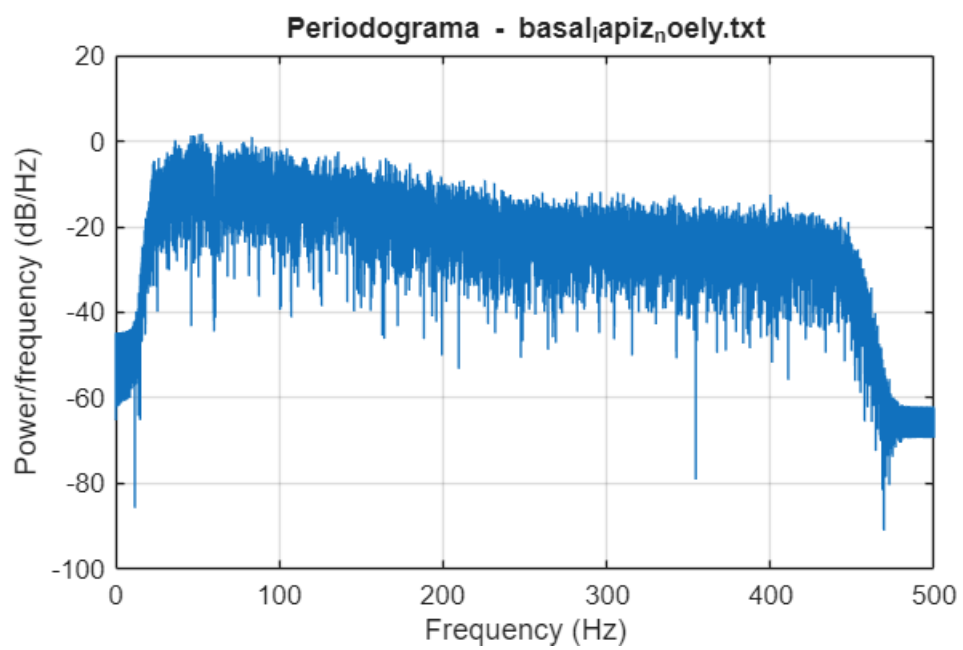
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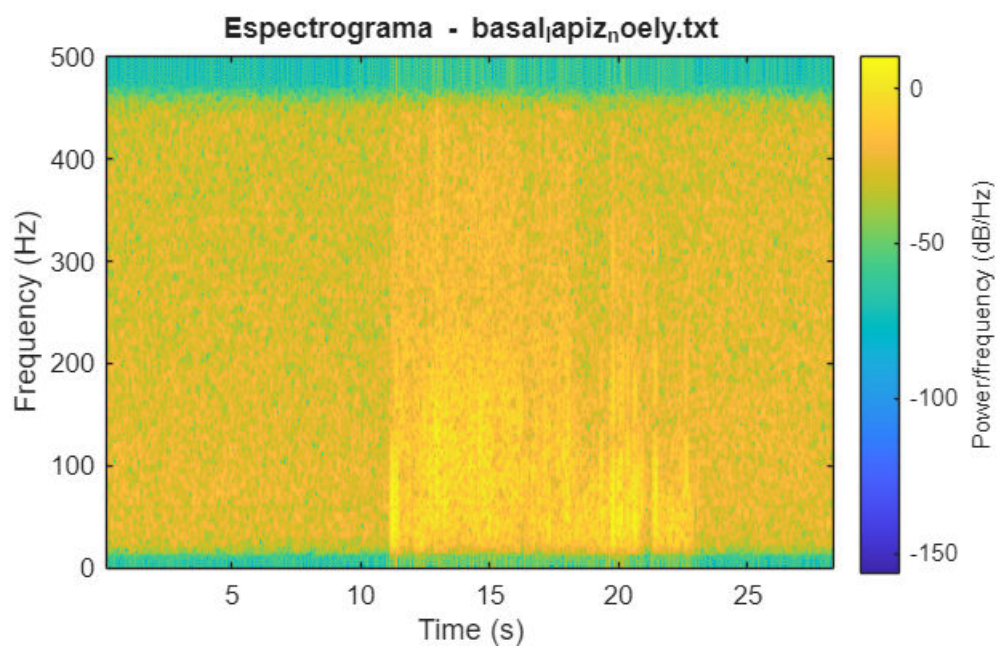


Procesando archivo 8 de 15...
basal_lapiz_noely.txt

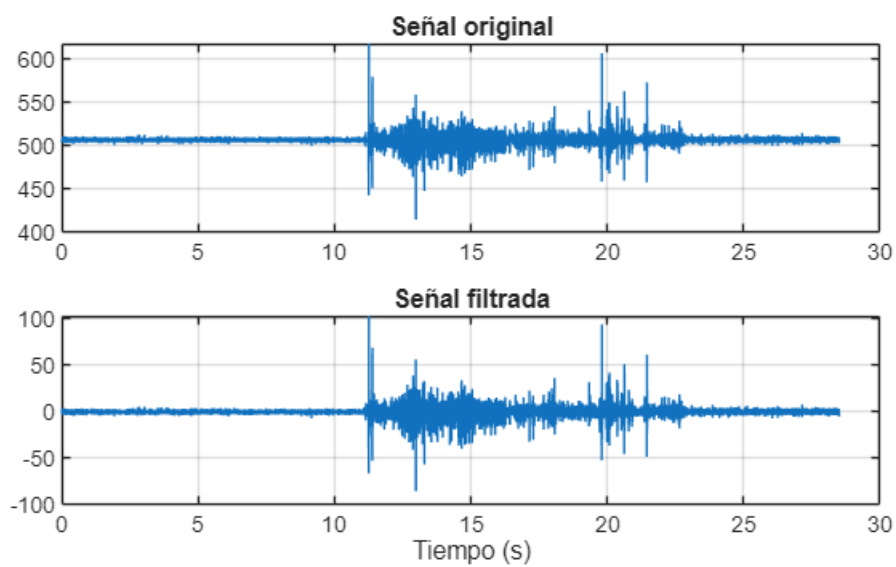




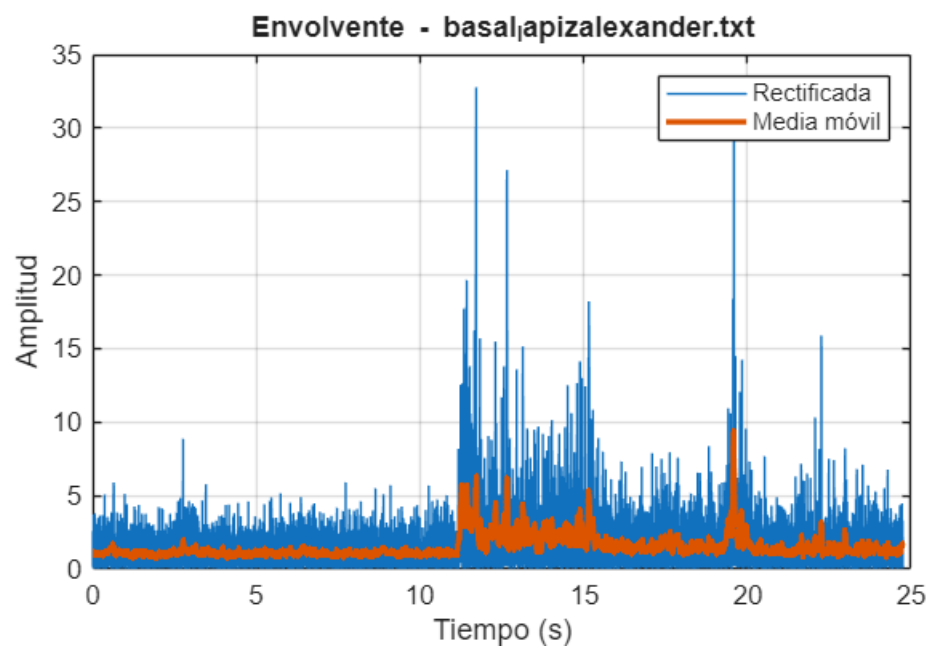
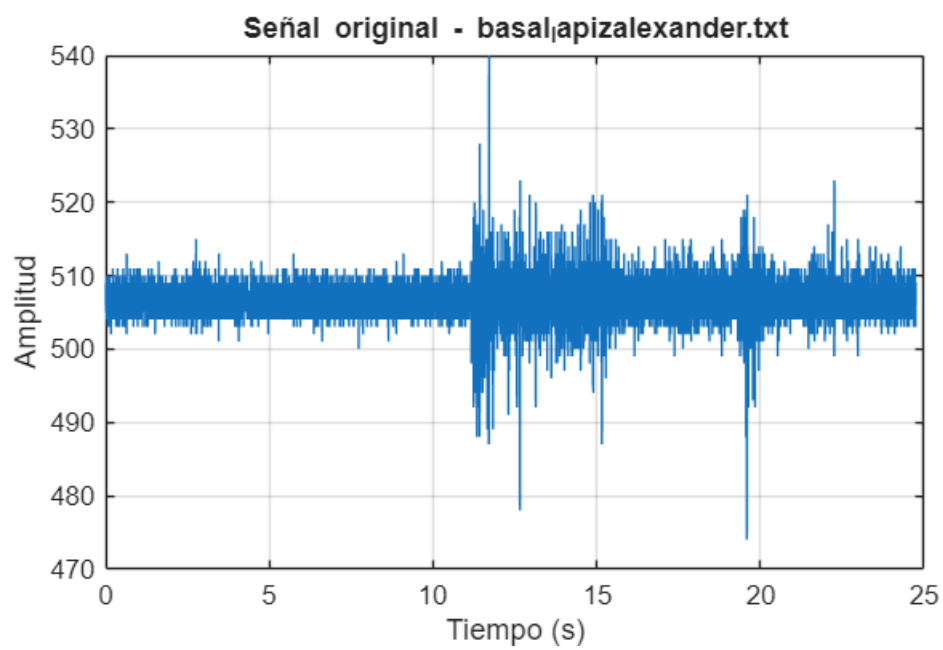


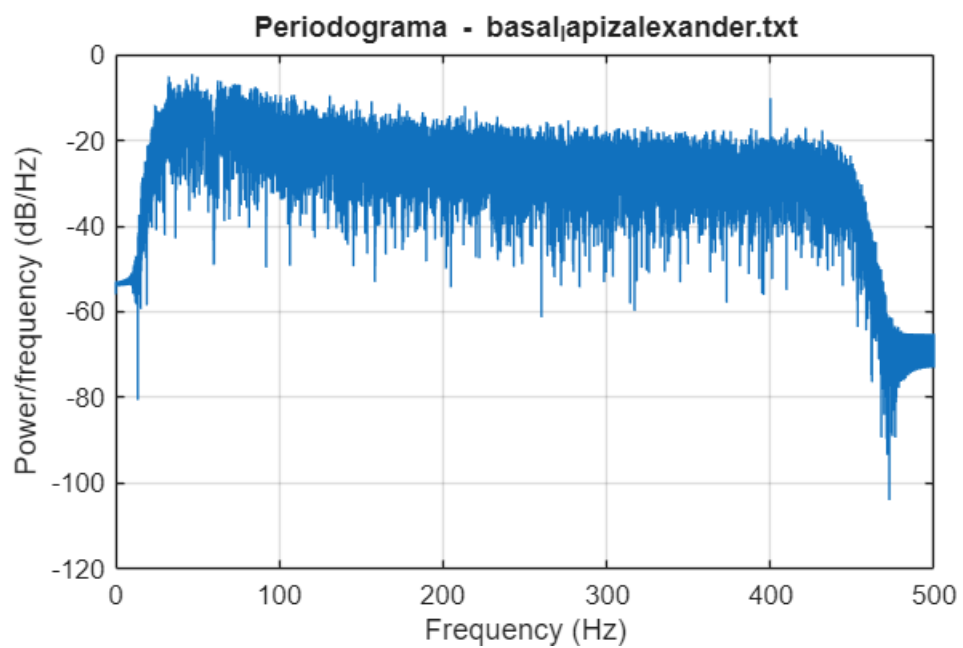
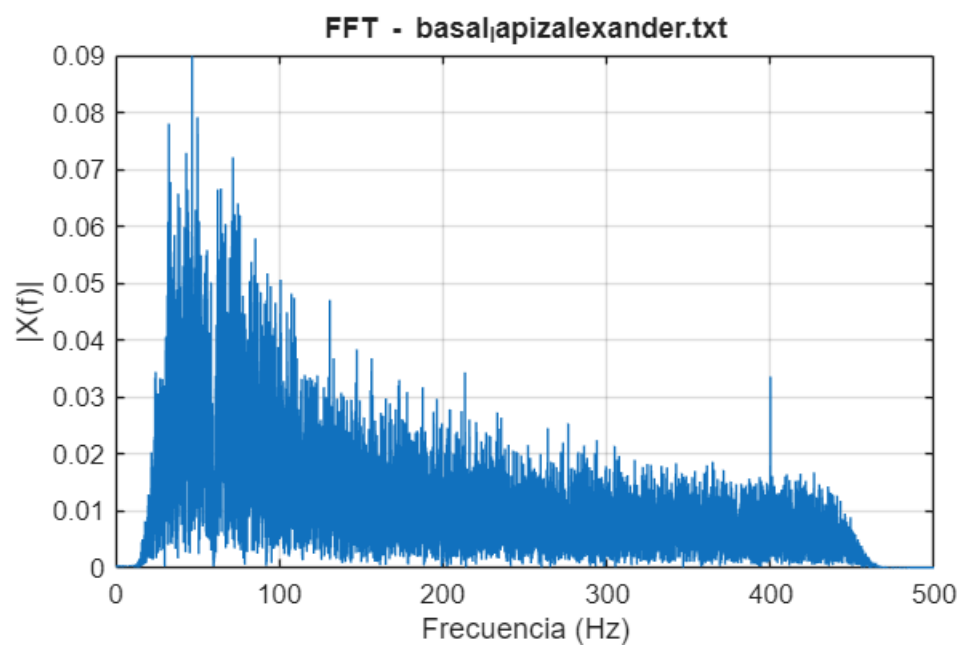


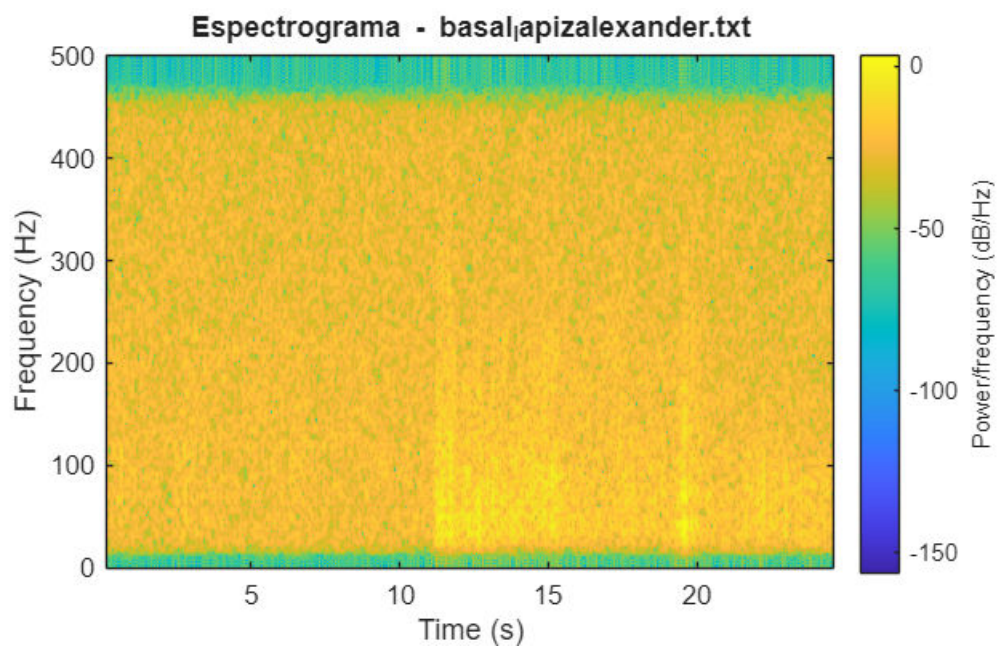
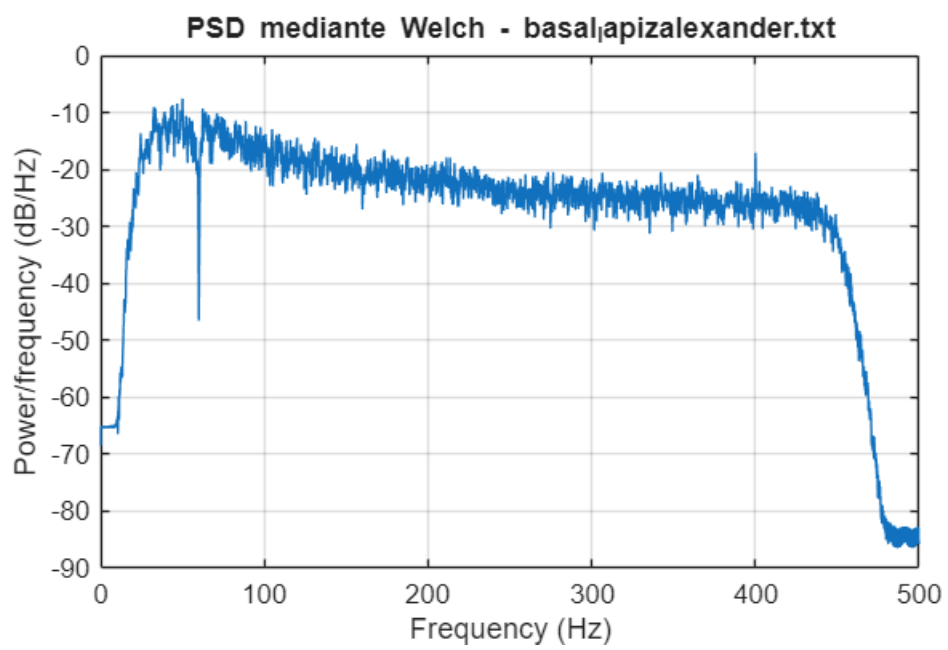
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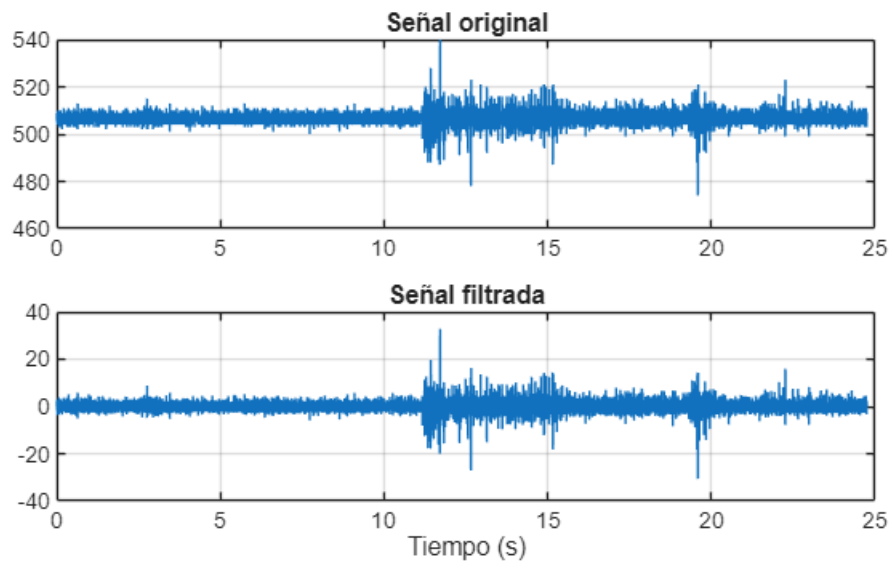
Procesando archivo 9 de 15...
basal_lapizalexander.txt



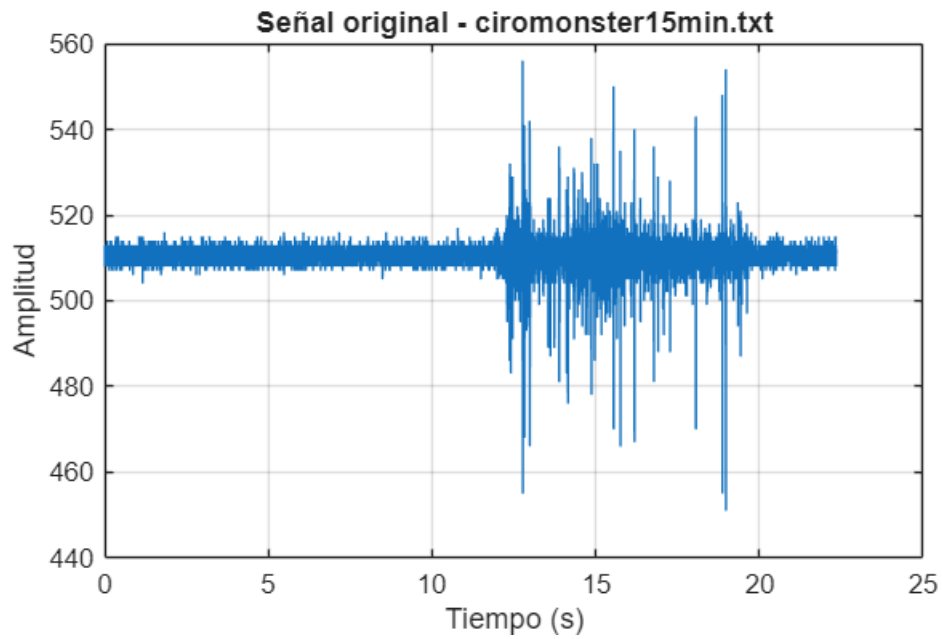


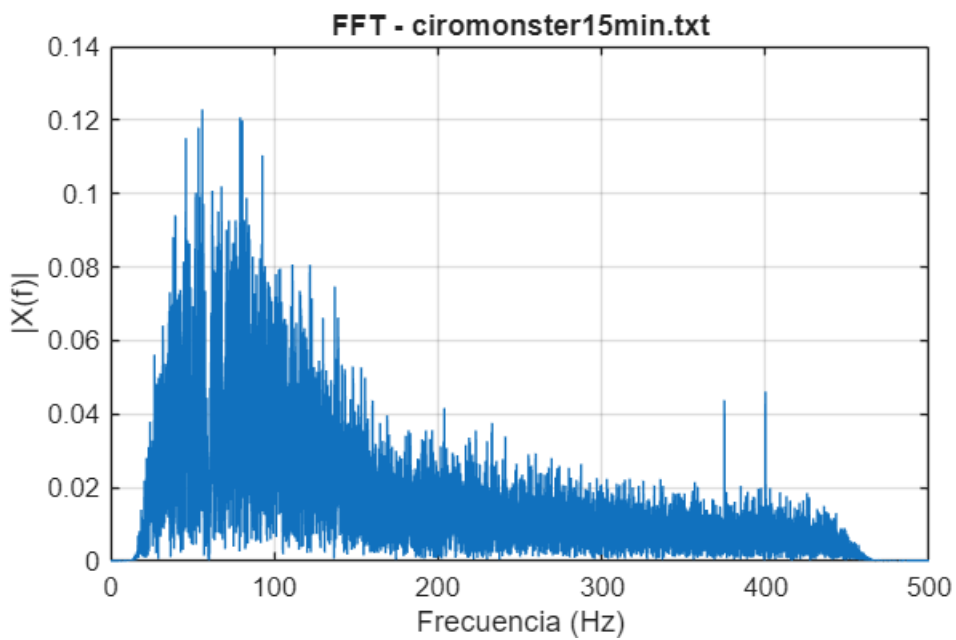
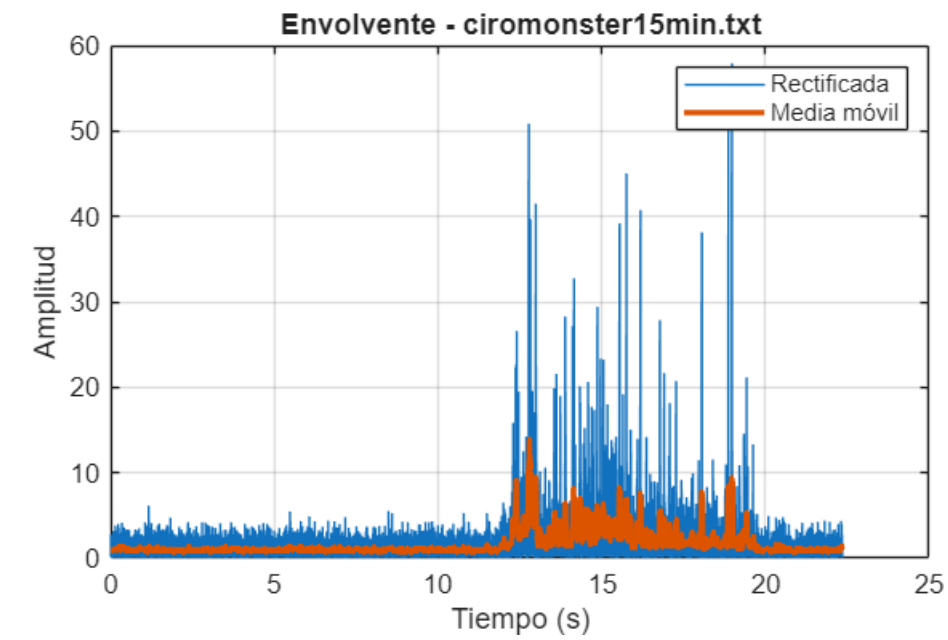


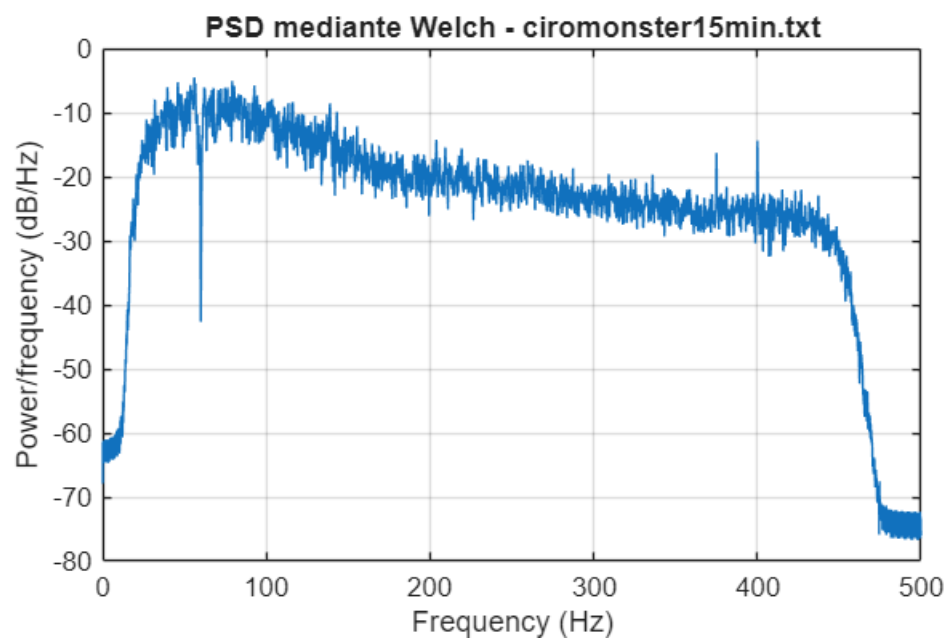
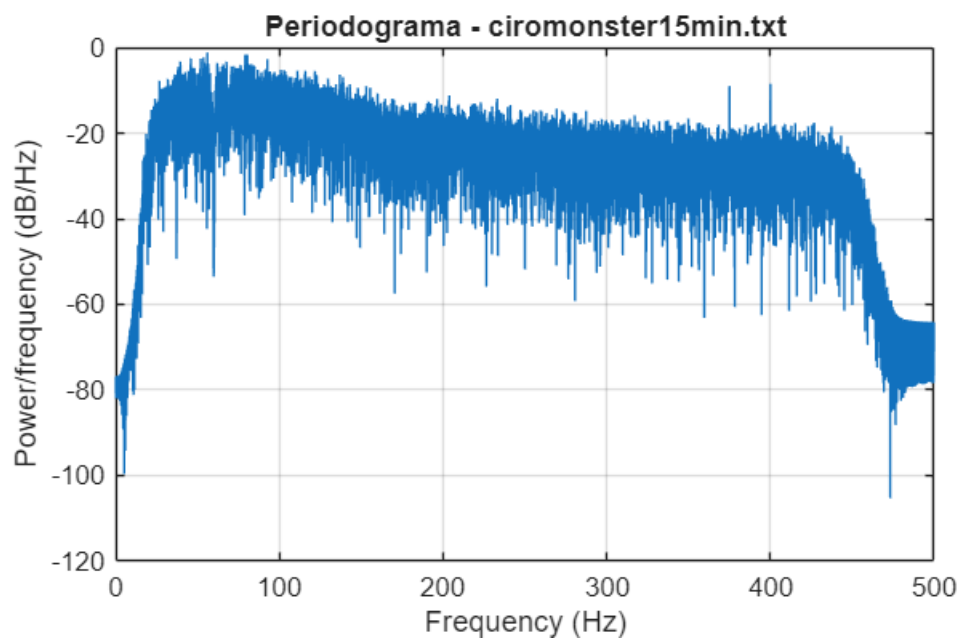
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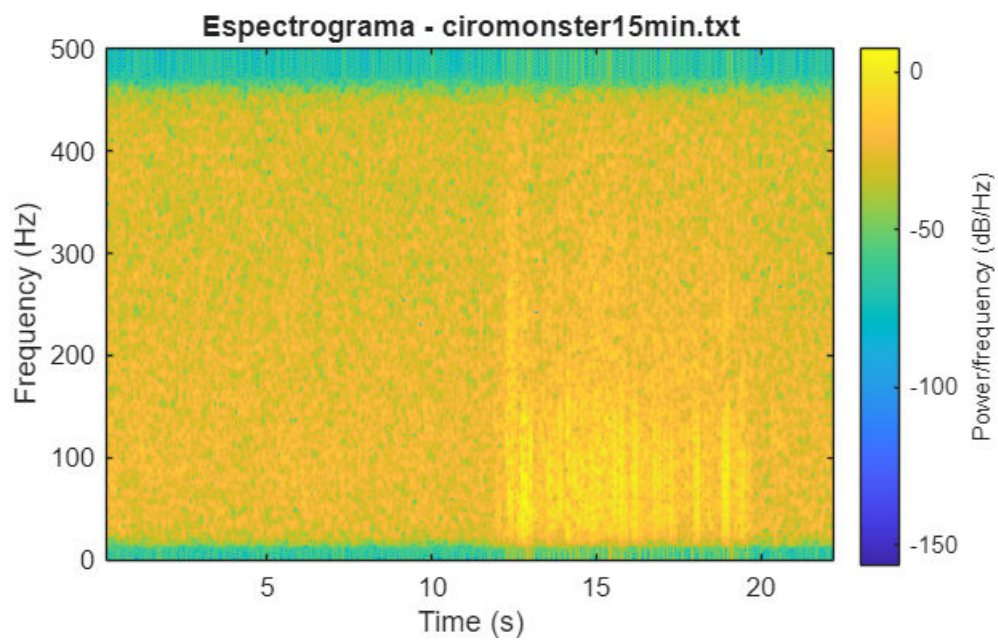


Procesando archivo 10 de 15...
ciromonster15min.txt

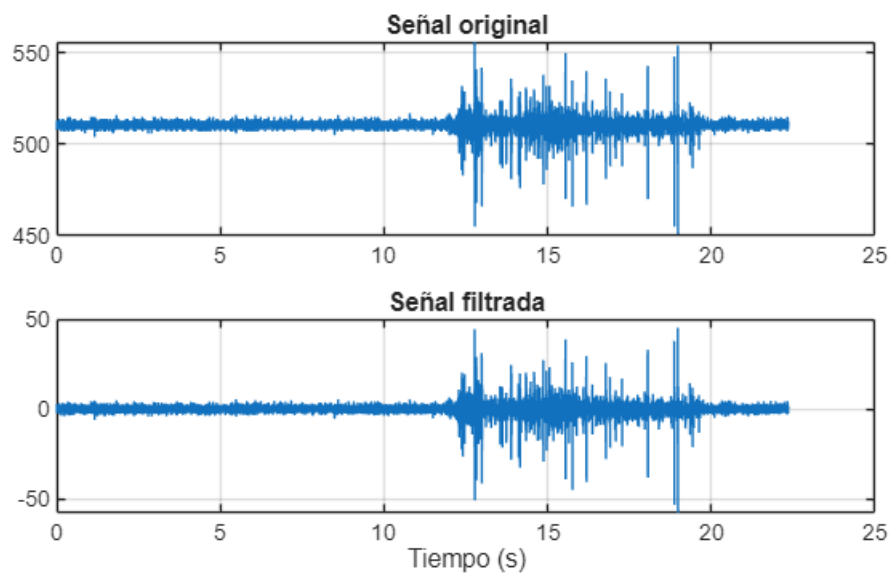




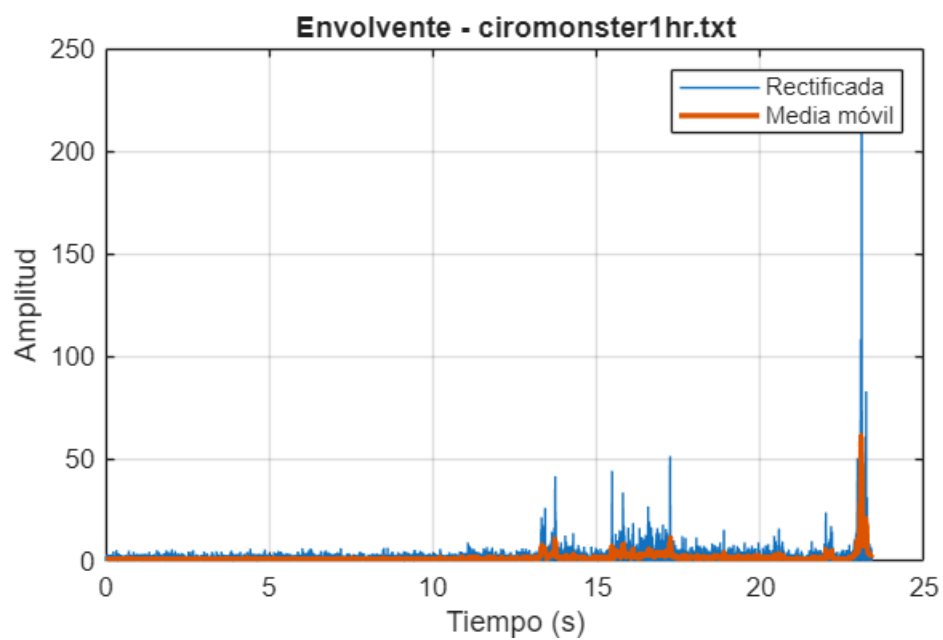
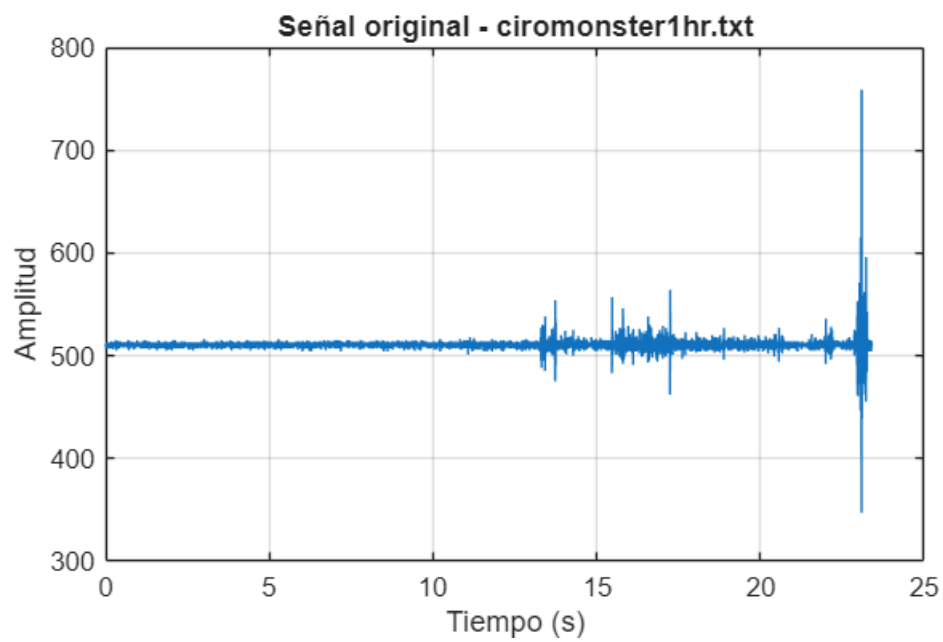


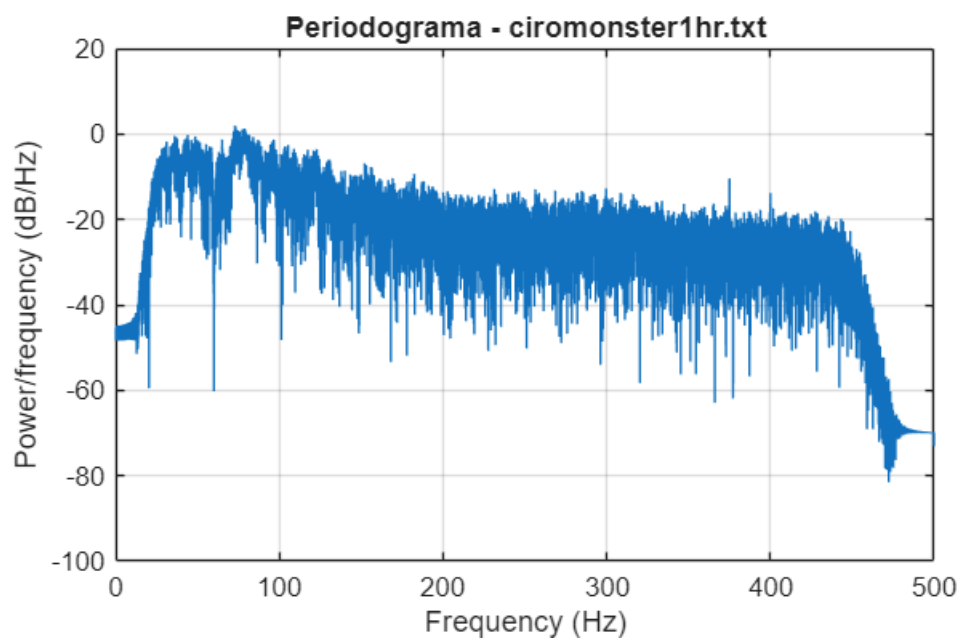
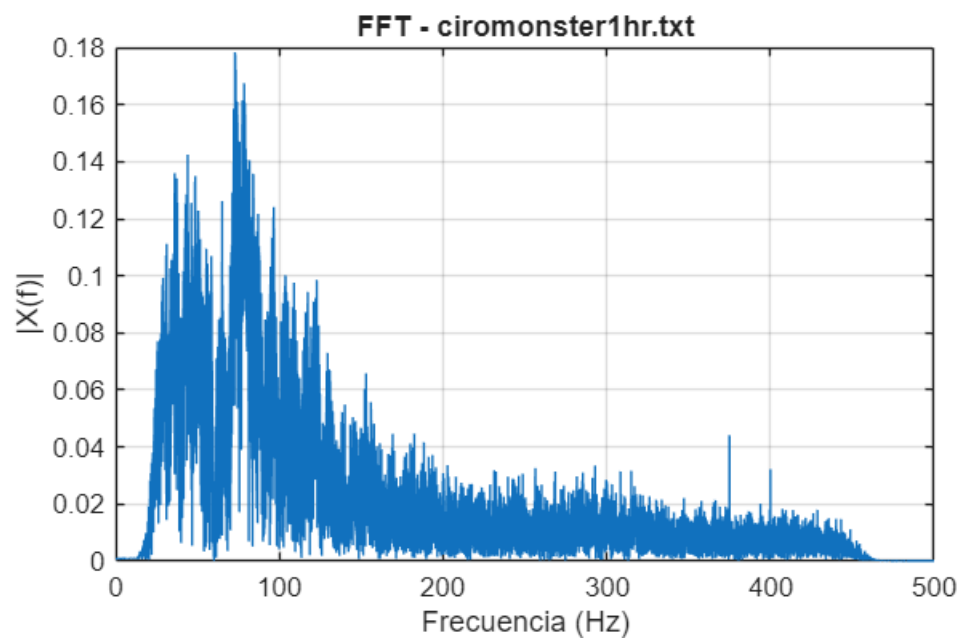


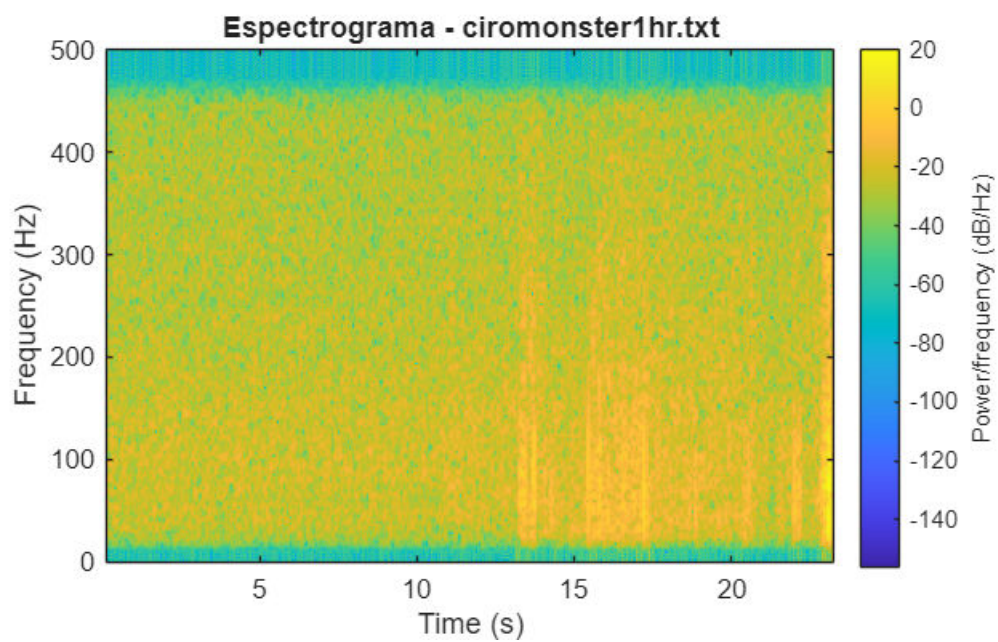
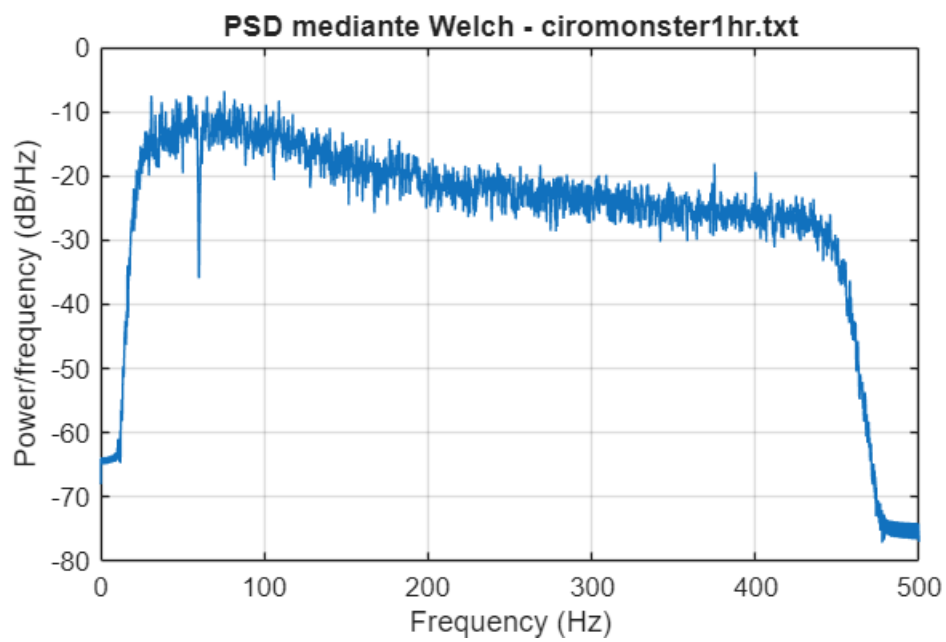
ciromonster15min.txt



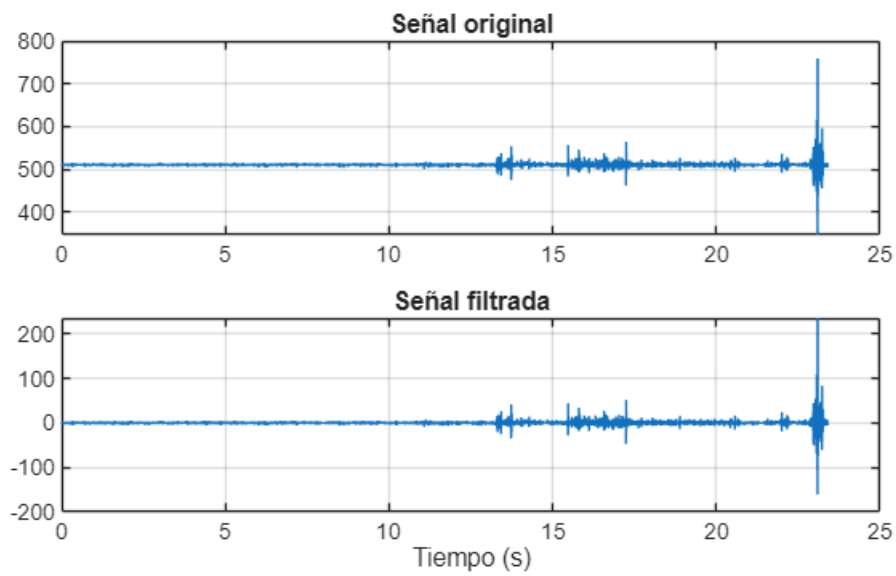
Procesando archivo 11 de 15...
ciromonster1hr.txt



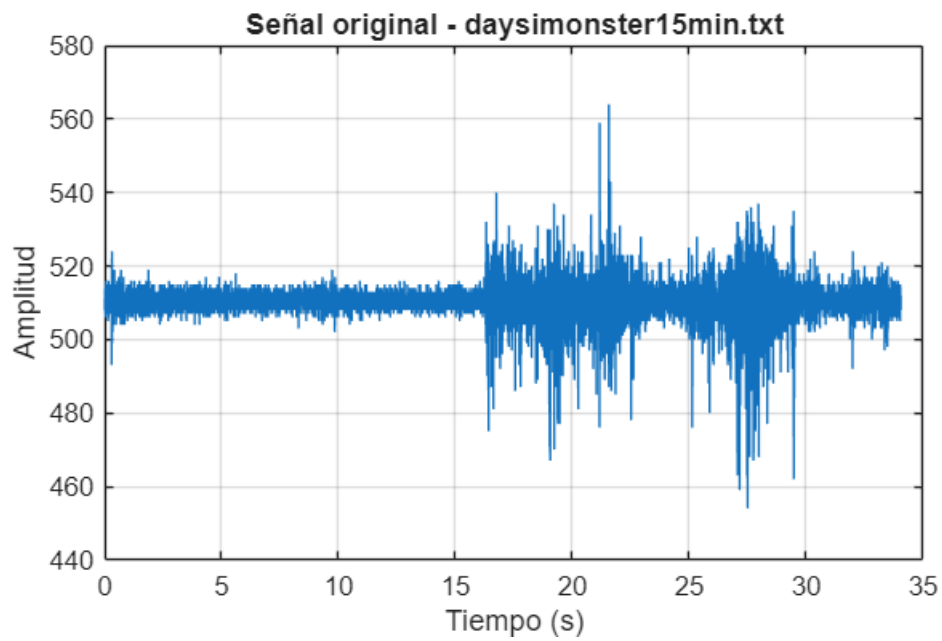


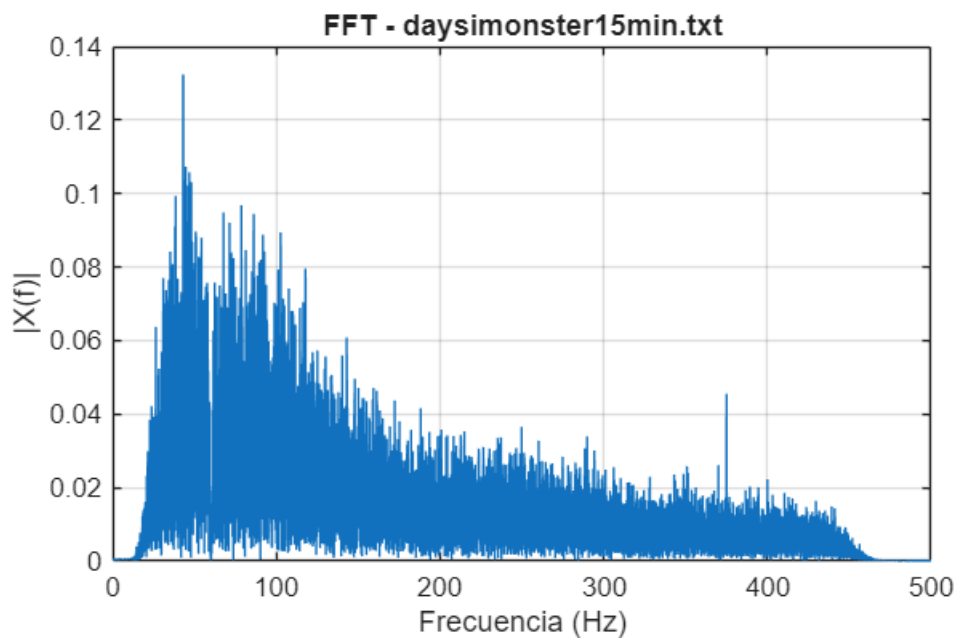
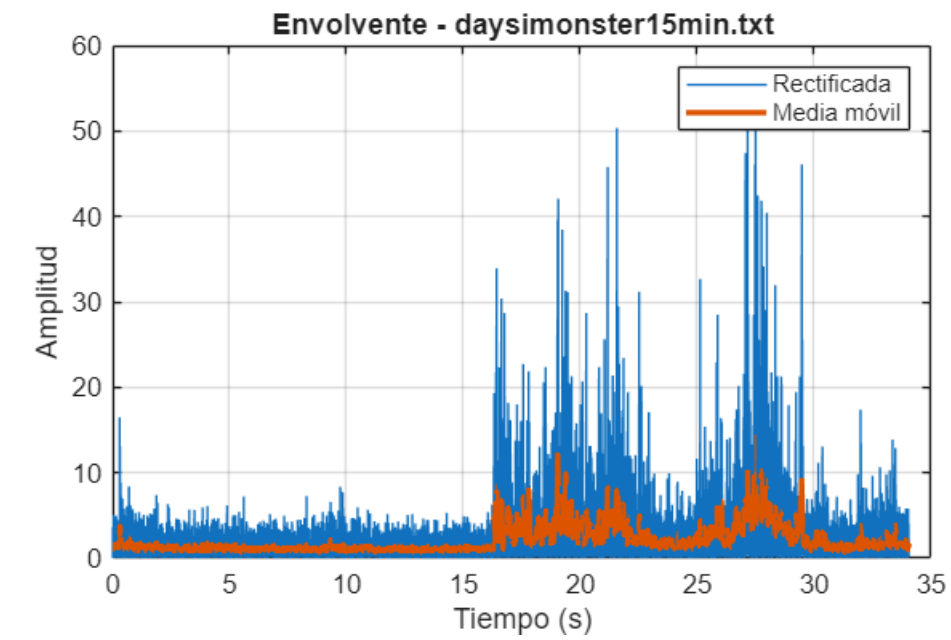


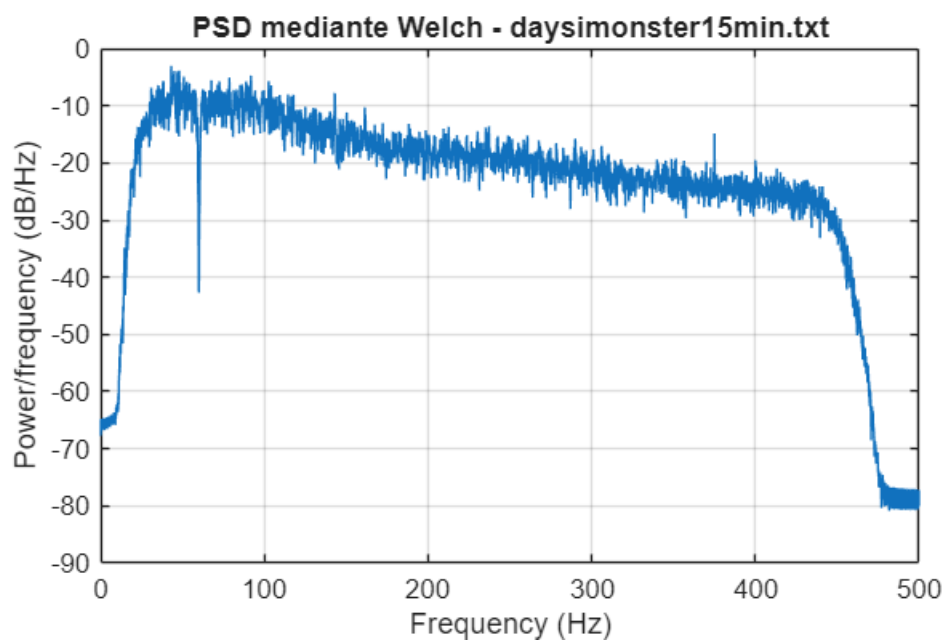
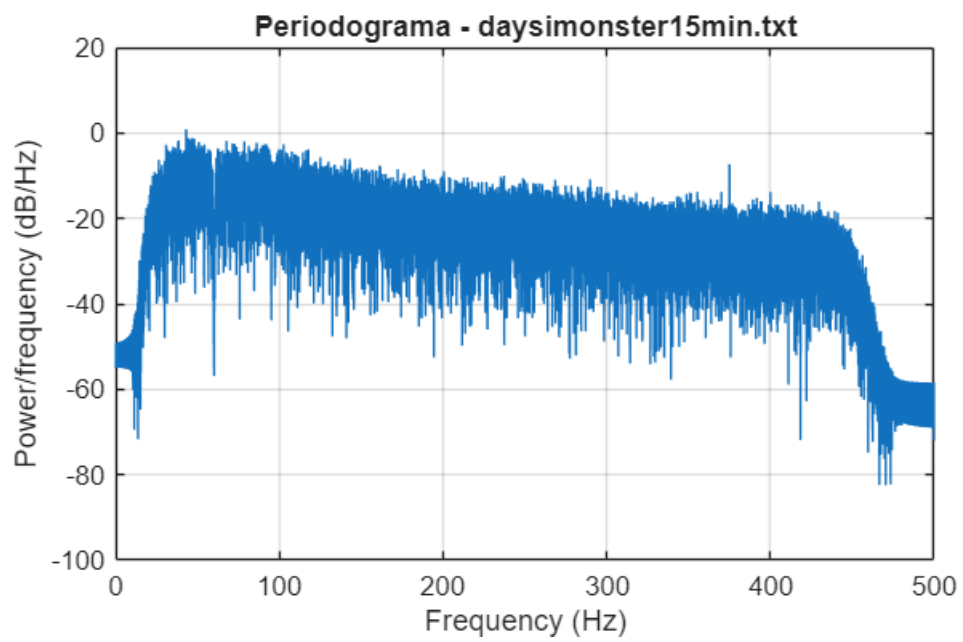
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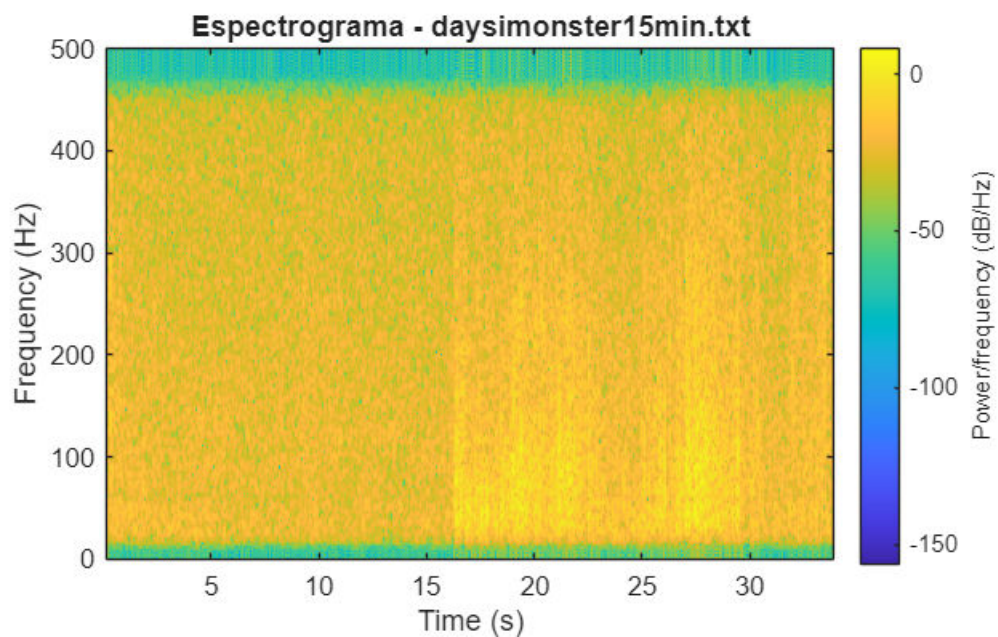


Procesando archivo 12 de 15...
daysimonster15min.txt

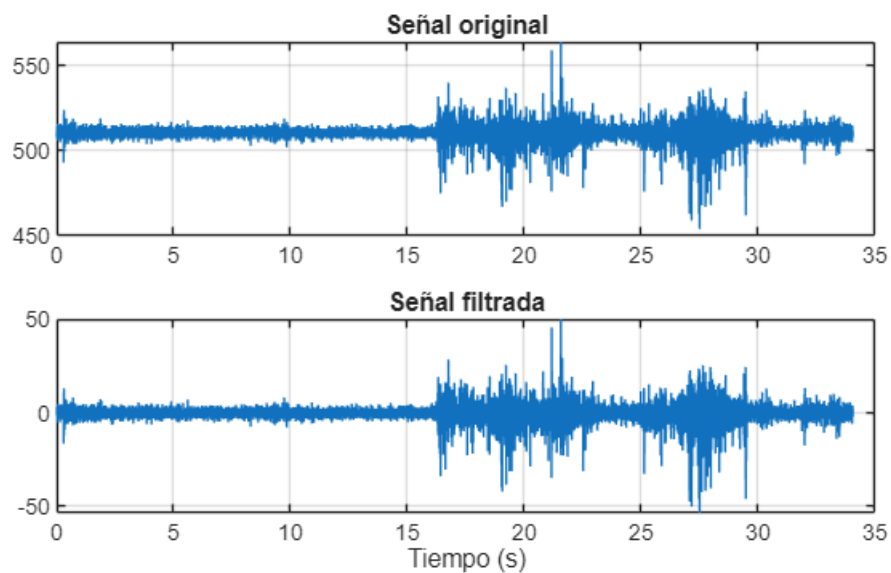




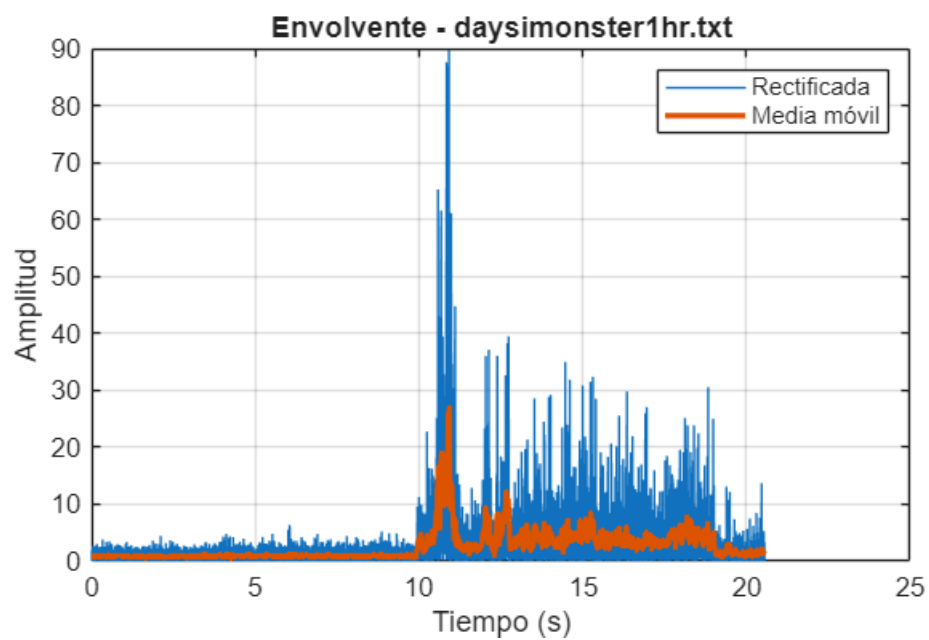
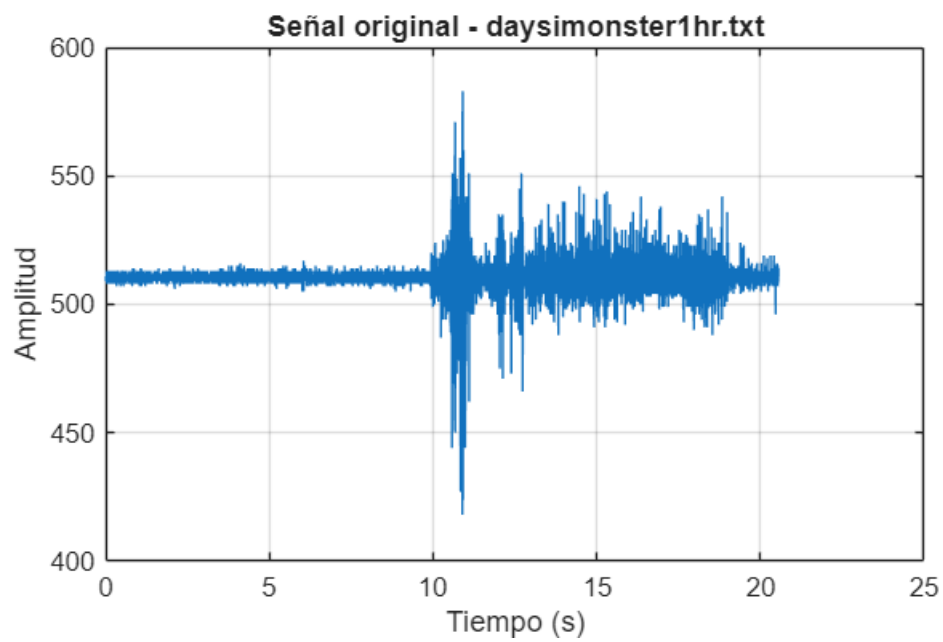


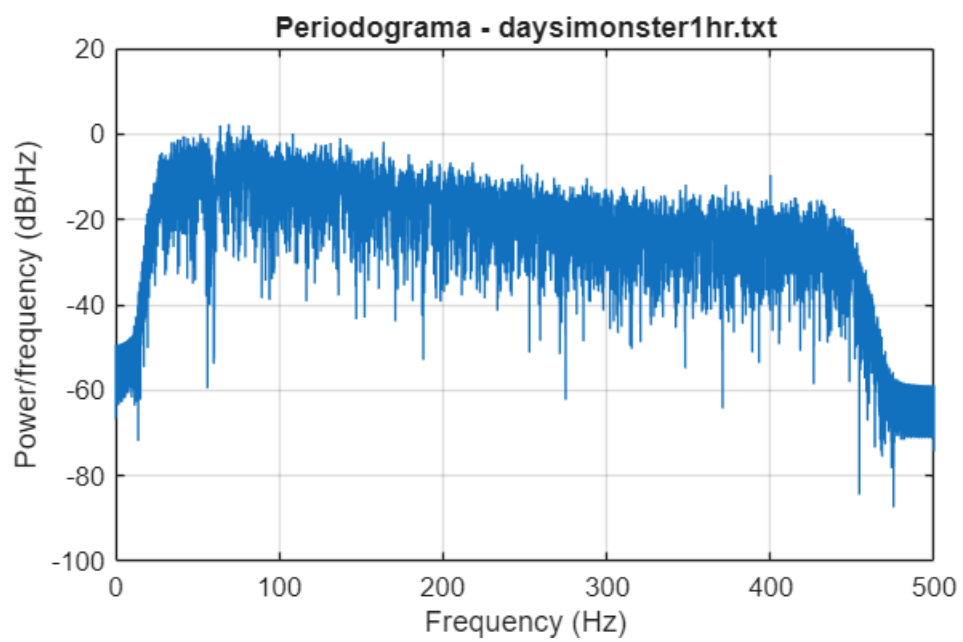
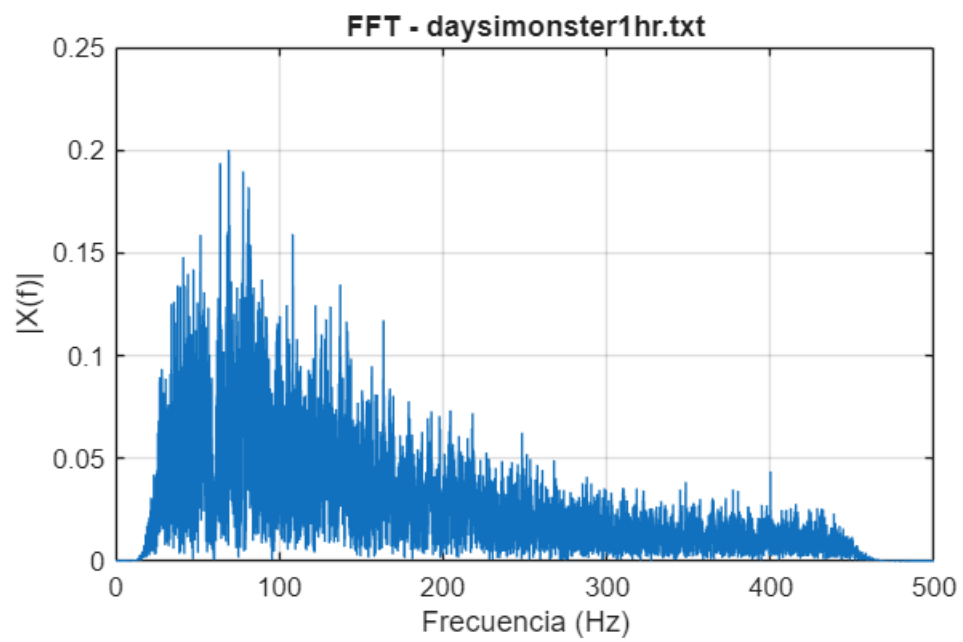


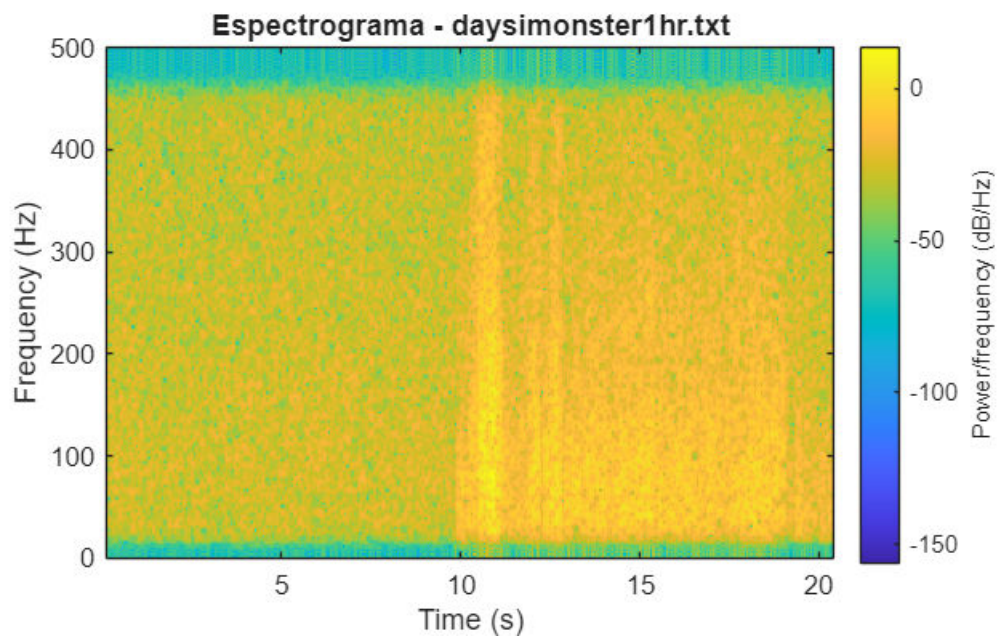
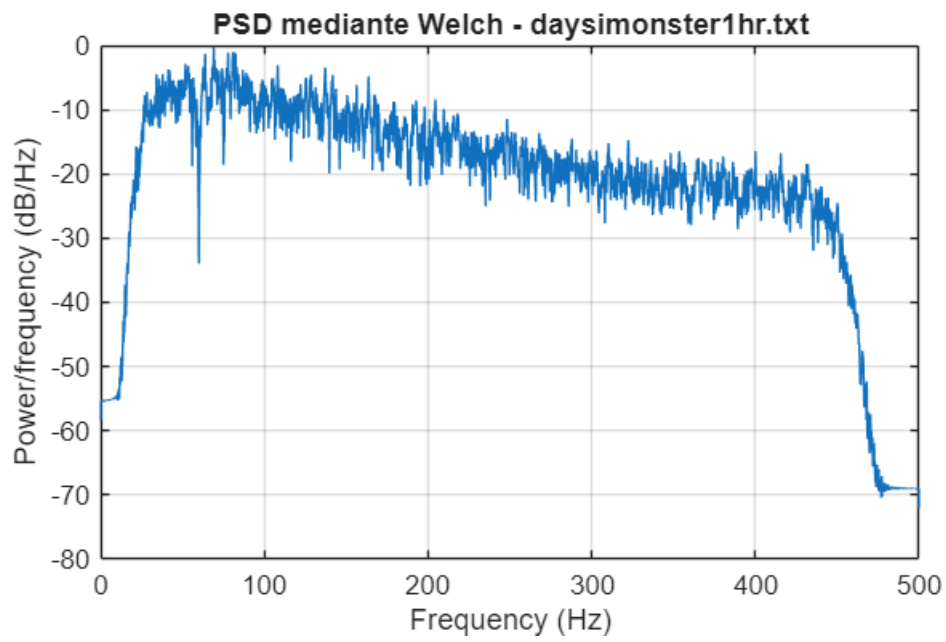
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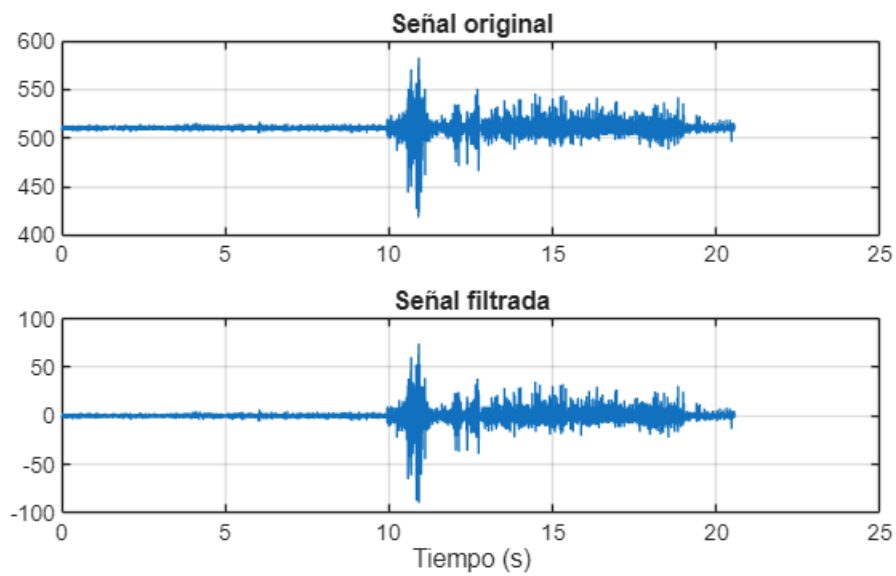
Procesando archivo 13 de 15...
daysimonster1hr.txt



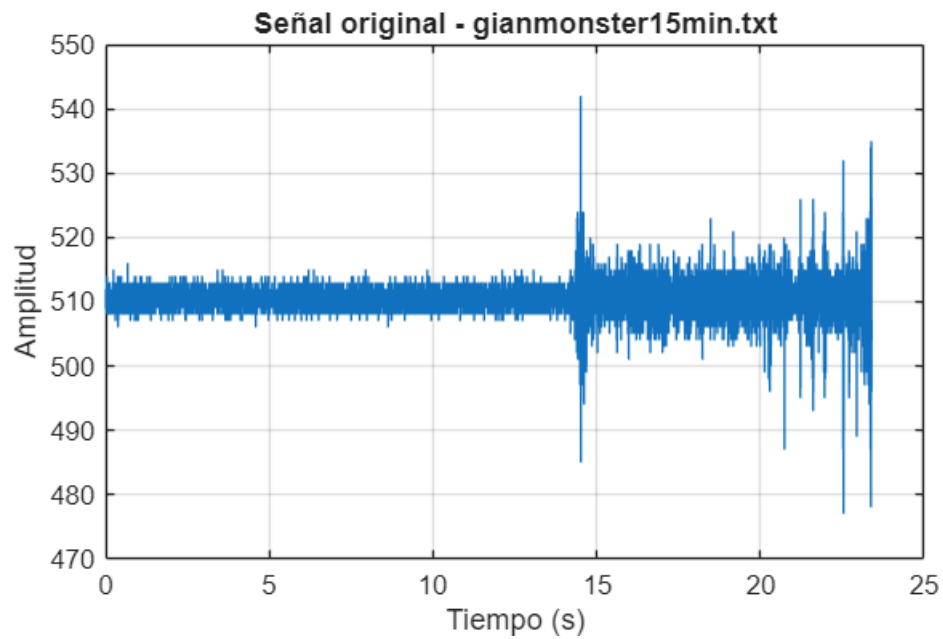


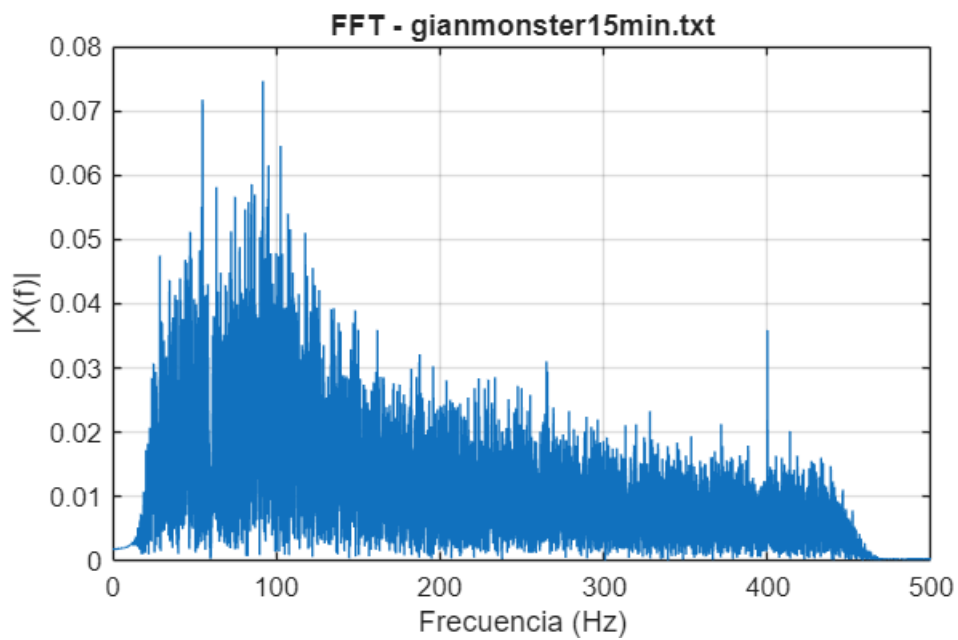
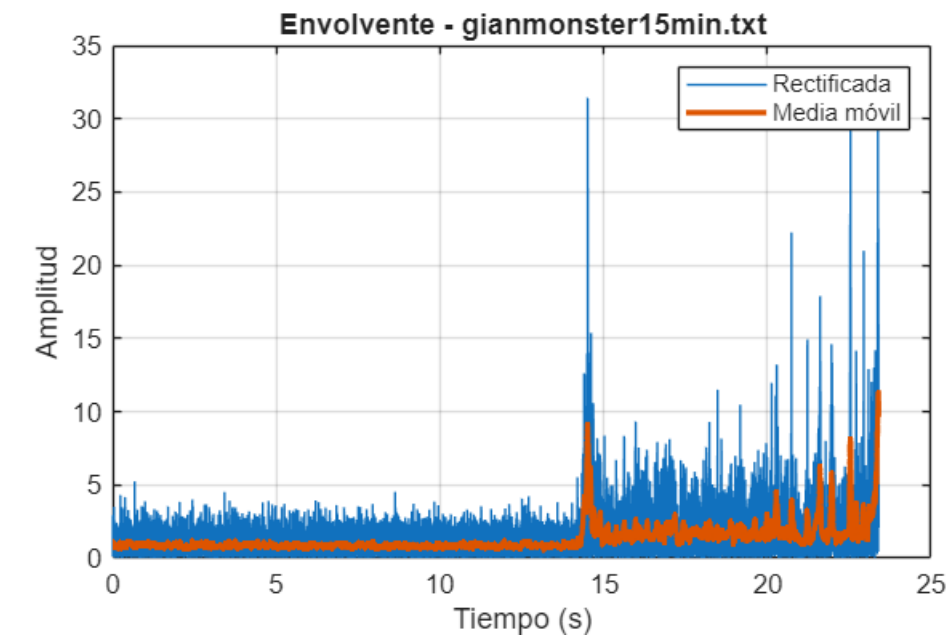


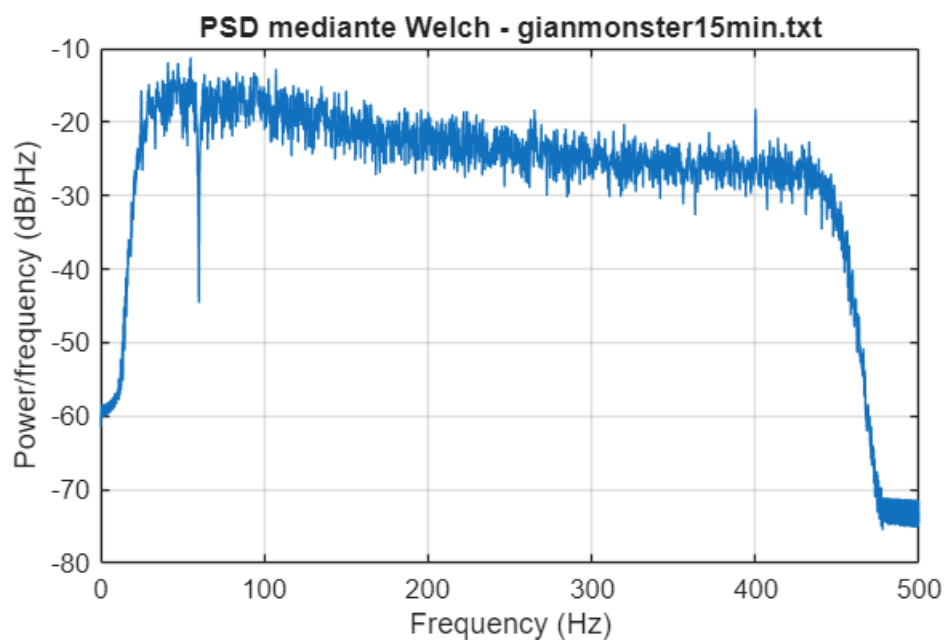
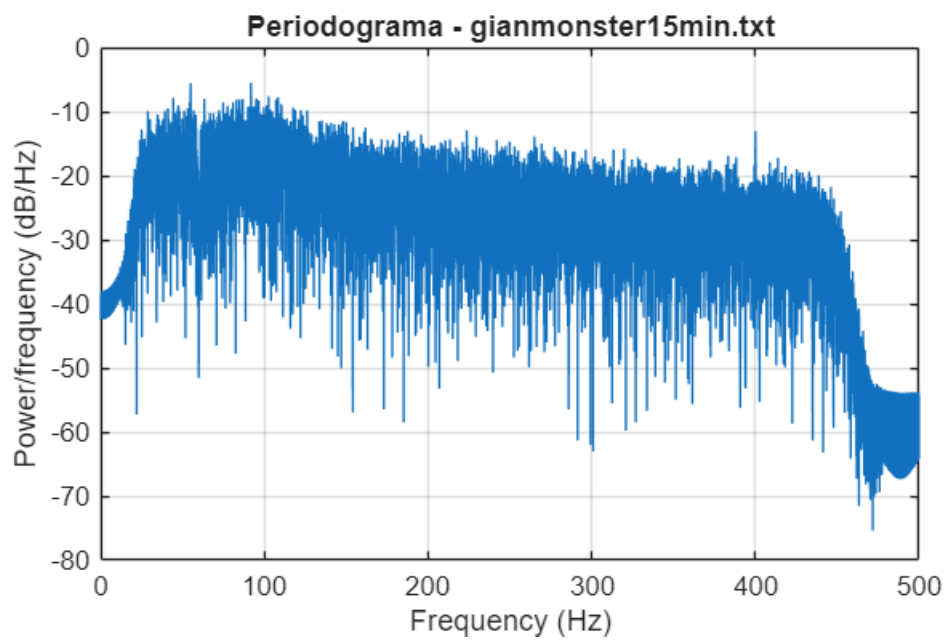
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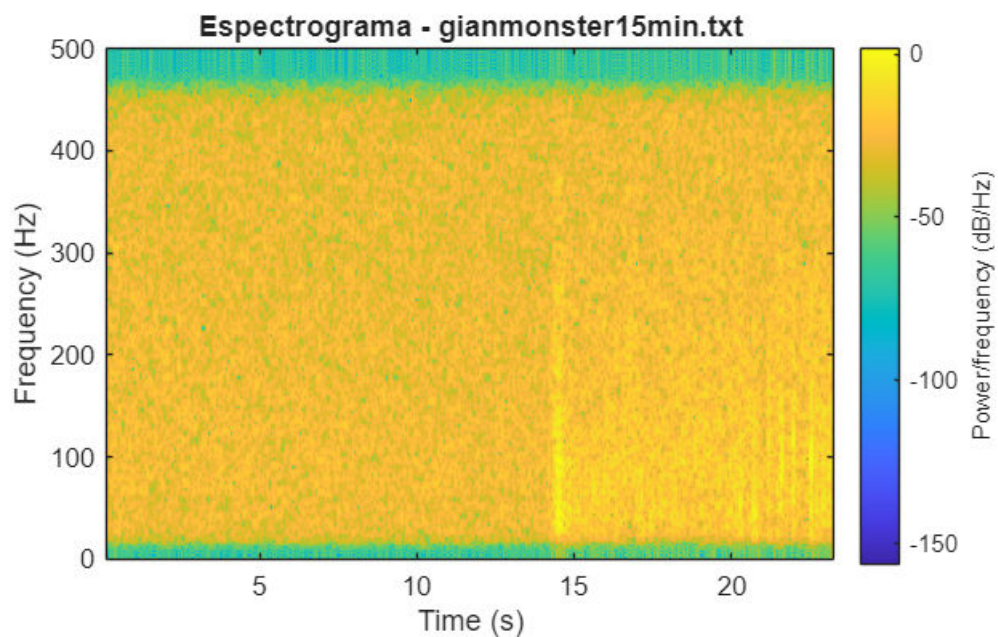


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gianmonster15min.txt

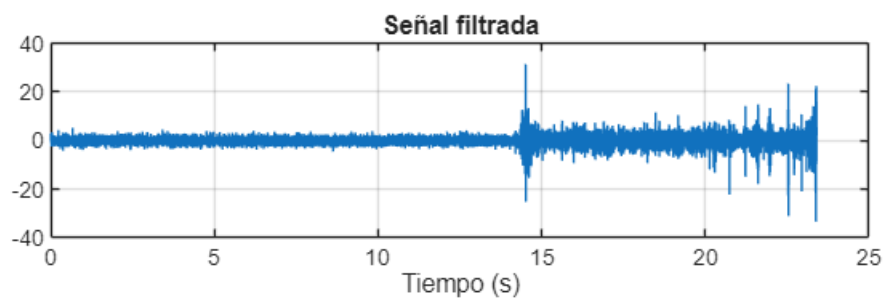
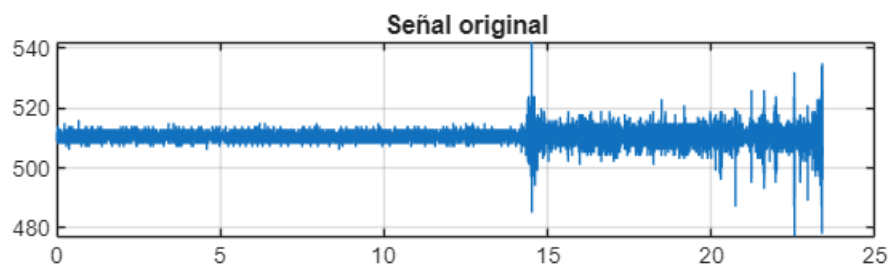




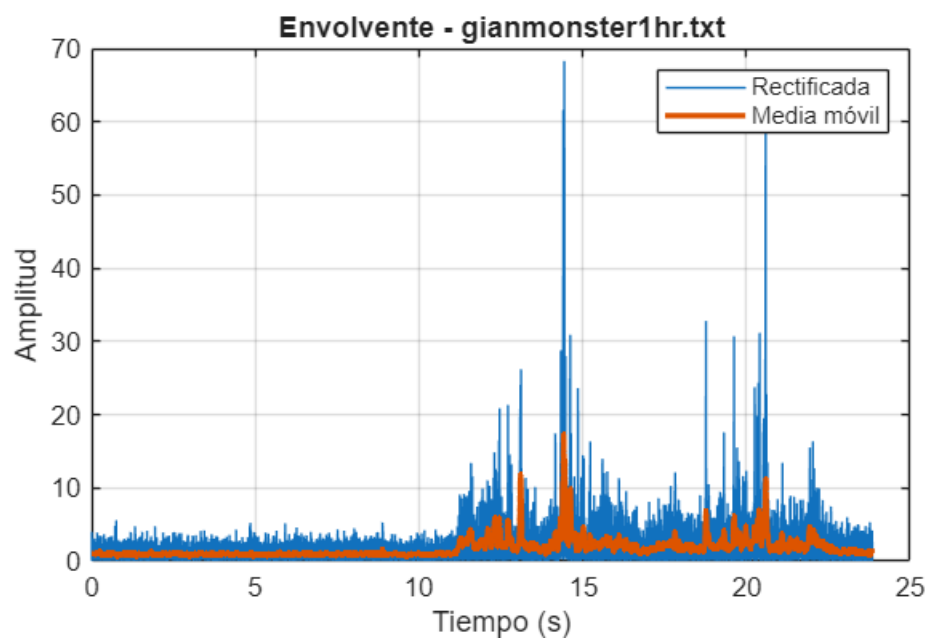
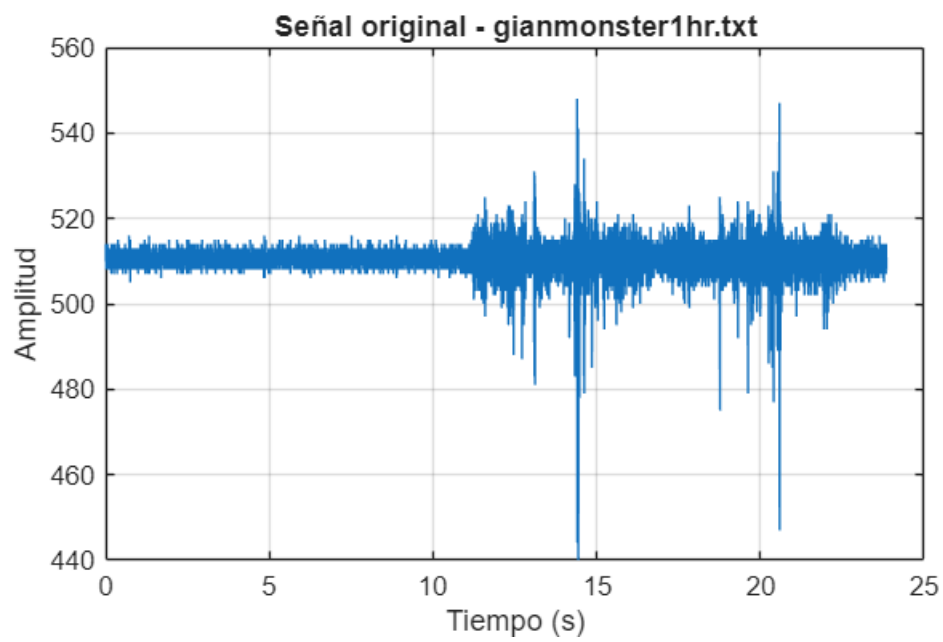


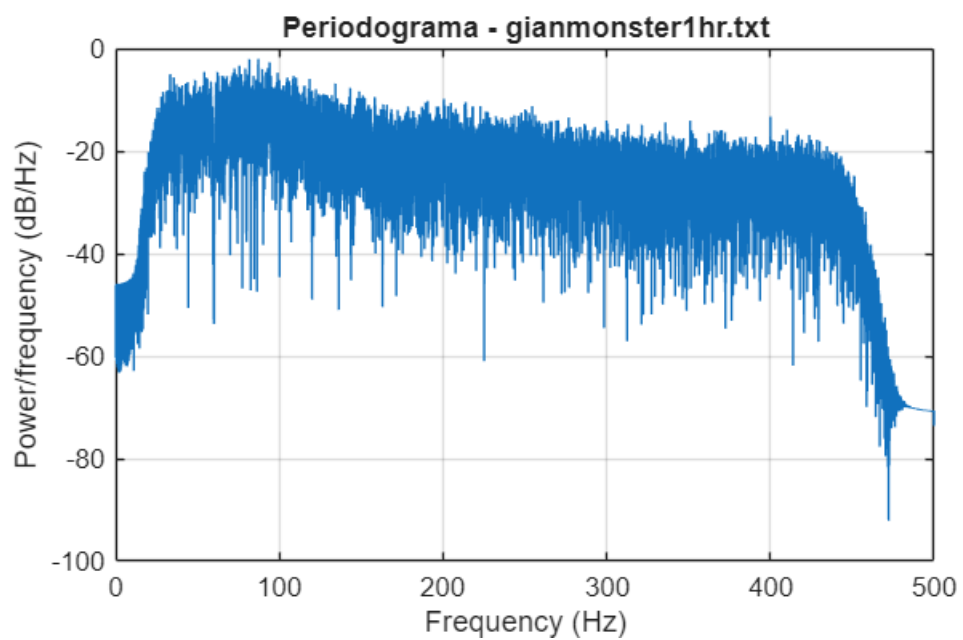
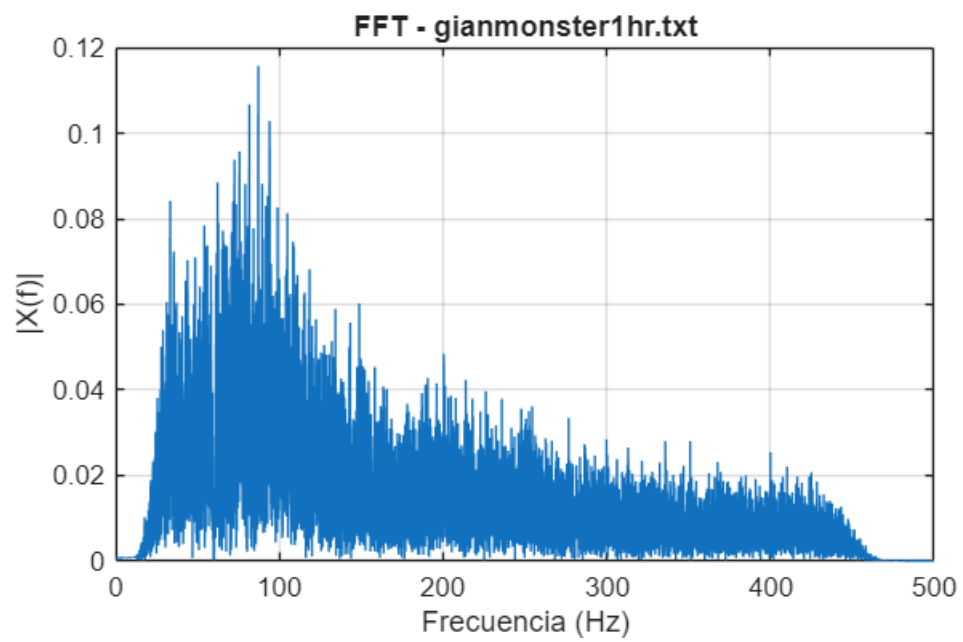


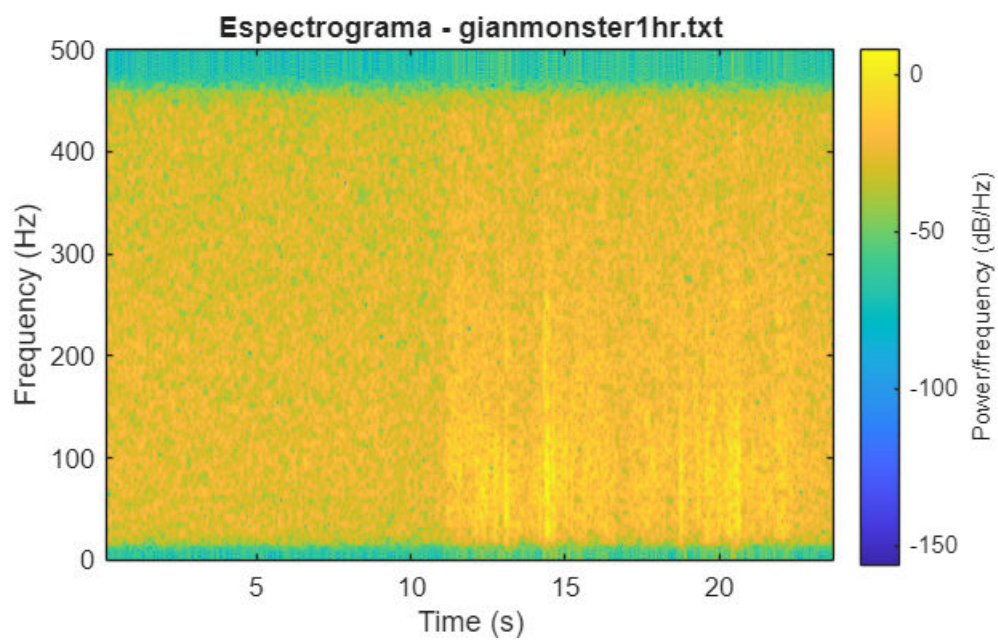
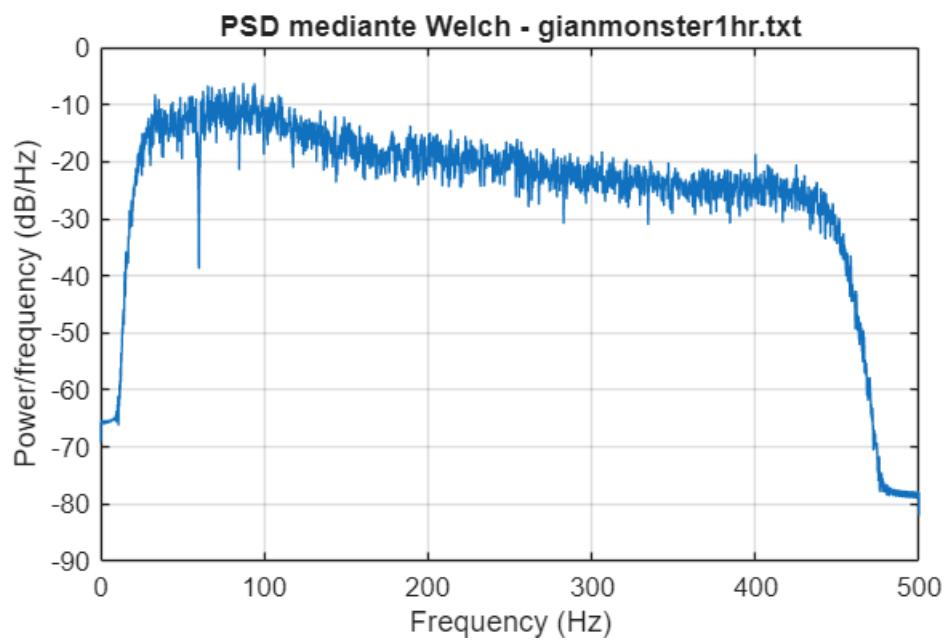
gianmonster15min.txt



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gianmonster1hr.txt







gianmonster1hr.txt

